No False Green: A Benchmark for Evidence-Aware Compliance Verdicts over Structured Employer Records

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Abstract

Automated systems increasingly present an organisation's regulatory obligations as a dashboard of satisfied and violated states. Their evaluation measures accuracy on individual determinations, which leaves one deployment-critical error unmeasured: reporting an obligation as *satisfied* when the evidence needed to assess it was never present. We call this a **false green**. It is worse than a false alarm in one specific way: a false alarm prompts the review that corrects it, while a false green removes the reason to review.

We introduce **No False Green**, a benchmark for evidence-aware compliance verdicts over structured employer records in United States employment-based immigration. A system receives an employer's data exports and a rulebook of citable obligations, and returns one of four verdicts per (obligation, entity) pair: `pass`, `fail`, `needs_evidence`, or `unmonitored`. The benchmark has two parts. A hand-authored anchor firm (221 pairs across eight regulatory regimes) carries labels from a rule engine, audited blind for whether the exports contain the evidence each label assumes. Generated firms carry labels fixed by construction: the generator chooses each pair's outcome and then writes records that decide it, including conforming cases placed exactly on the legal boundary. Trivial baselines show that no single metric suffices. Answering `pass` everywhere raises no false alarm while calling every unassessable obligation satisfied, and abstaining everywhere scores well on abstention while detecting nothing.

We evaluate nine language models, including Google's frontier Gemini 3.1 Pro, on 140 pairs from the anchor firm and 252 from six generated firms. Offered an abstention, models report 0.042 to 0.153 of the anchor firm's unassessable obligations as satisfied while detecting 0.050 to 0.500 of its violations. A dashboard-style schema permitting only `pass` and `fail` raises the false-green rate in every model, to 0.131 to 0.589. Two further conditions show that the cause depends on the model. Removing only the abstention verdict changes little, because models move to another non-committal verdict. With no non-committal verdict left, a weaker model calls most unassessable obligations satisfied, while the frontier model holds as long as `pass` is defined as requiring evidence. On the anchor firm, false greens concentrate where an export carries a populated column next to the missing evidence; on clean generated exports, the definition of `pass` makes no difference.

Every instance is synthetic, so the benchmark is released without the confidentiality constraints that keep comparable compliance data unpublishable.

1. Introduction

A compliance dashboard makes a claim about the world with every cell it renders. When a cell is green, it asserts that a specific legal obligation, attached to a specific worker or filing, has been assessed and found satisfied. The assertion has a precondition that the rendering does not express: that the evidence needed to perform the assessment was present.

When the precondition fails, a system has three options. It can say so. It can decline to render the cell. Or it can render the cell green. The third option is the one this paper is about, and it is attractive to build for reasons that have nothing to do with correctness: a dashboard of mostly-green cells looks like a working product, a mostly-grey one looks broken, and the customer cannot distinguish "assessed and fine" from "unassessed" without reading the underlying records themselves, which is the labour the product exists to remove.

We call the third option a **false green** and make three claims about it.

It is not symmetric with a false alarm. A false alarm consumes reviewer time and is extinguished by the review it triggers. A false green consumes nothing and triggers nothing. It is self-concealing: the only signal that would surface it is the review that it suppresses. In immigration compliance the cost is concrete. Form I-9 paperwork violations are penalised per form (8 C.F.R. 274a.10(b)(2)) and LCA violations per violation (20 C.F.R. 655.810(b)). A Form I-9 inspection gives the employer at least three business days' notice (8 C.F.R. 274a.2(b)(2)(ii)), and technical or procedural failures can be corrected within ten business days of notification (8 U.S.C. 1324a(b)(6)(B)), but substantive violations cannot, and USCIS FDNS site visits are generally unannounced. A gap found before an inspection can often be remedied; the same gap found during one becomes a finding. A false green is what keeps it from being found before.

It is invisible to pass/fail scoring. An evaluation that labels every pair either compliant or violated has no category for "the evidence was not there", so missing evidence is scored as compliant, and a system that answers `pass` whenever evidence is absent is never penalised. Violations are also a minority class, so on compliant cases the same system scores perfectly. Section 8 shows the second point concretely: our `always_pass` baseline raises no false alarm in any of the 24 firms that carry conforming controls.

It is barely measured. Benchmarks in legal NLP evaluate reasoning and classification over regulatory text and case law (Guha et al., 2023). Most abstention benchmarks evaluate free-text question answering. Recent work has begun to score evidence sufficiency and the overclaiming direction directly, in retrieval-augmented generation (Joren et al., 2024), claim verification over databases (Theologitis et al., 2026), and agent governance and authorisation-limited evidence (Solozobov, 2026; Tallam, 2026). None of it evaluates a system reading a third-party employer's structured records and deciding, per legal obligation, whether the records support a verdict at all, in a domain where correct abstention is a large share of the task by design. Section 2 sets out how close each comes.

We make four contributions.

1. **A task.** Verdicts over (obligation, entity) pairs with four states, where `needs_evidence` is a first-class output with its own correctness condition rather than a refusal to participate (Section 4).
2. **A benchmark.** A hand-authored anchor firm with 221 audited expectations across eight regimes; a seeded generator producing arbitrarily many (if deliberately narrow) firms whose labels are fixed by construction, including conforming cases on the legal boundary; and a rulebook of 194 obligations, each with its regulatory citation (Section 5).
3. **A metric suite** centred on the false-green rate, with baselines demonstrating that no single metric in the suite is sufficient alone (Sections 6 and 8).
4. **An empirical study** of nine language models, including a current frontier model, on both parts of the benchmark. It isolates the effect of removing the abstention option from the output schema, with repeated runs and a one-variable ablation (Sections 8 and 9).

The benchmark is fully synthetic, which is the reason it can exist. Compliance data describing real employers is subject to attorney-client privilege and employee privacy obligations that make publication impossible, which is why the gap this benchmark fills has remained open.

Terms. *Form I-9*: the employment-eligibility verification form every US employer completes for each hire; *Section 1* is completed by the employee, *Section 2* by the employer after examining documents, and *Supplement B* records reverification when temporary work authorisation expires. *E-Verify*: the federal electronic eligibility check; a *TNC* (tentative nonconfirmation, now called a mismatch) requires notice and a response window. *H-1B*: the temporary specialty-occupation visa. *LCA*: the Labor Condition Application an employer files with the Department of Labor before an H-1B petition, attesting to wages and conditions; its *PAF* (public access file) must hold supporting documents. *Prevailing wage* and *actual wage*: the two floors of the LCA required wage, which is the greater of them. *SOC*: Standard Occupational Classification code. *I-94*: the arrival-departure record. *EAD*: employment authorisation document. *RFE* and *NOID*: USCIS requests for evidence and notices of intent to deny, each with a response deadline.

FDNS: the USCIS Fraud Detection and National Security directorate, which conducts site visits. *AAO*: the USCIS Administrative Appeals Office.

2. Related work

Several literatures touch this problem. None occupies quite the position this benchmark takes, but three recent benchmarks come closer than a reader of the introduction might expect, and we say where. In each case the difference lies in what the system is given, what it must return, and how the label is established.

Legal and regulatory NLP benchmarks. Work in this area evaluates reasoning and classification over *legal text*. LegalBench assembles 162 tasks spanning six types of legal reasoning, designed and hand-crafted by legal professionals (Guha et al., 2023). LexGLUE standardises a collection of legal language-understanding datasets (Chalkidis et al., 2022); CaseHOLD poses over 53,000 multiple-choice questions identifying the holding of a cited case (Zheng et al., 2021); CUAD asks models to highlight salient portions of contracts, with over 13,000 annotations from legal experts (Hendrycks et al., 2021); and LawBench tests legal knowledge at the levels of memorisation, understanding and application (Fei et al., 2024). A system is given a legal question or a document and asked to answer, classify, or retrieve. Our input is an organisation's structured operational records, and our output is a verdict per obligation about that organisation. A model that scores well on legal reasoning has demonstrated nothing about whether it will report an unassessable duty as satisfied, because these tasks do not ask it to decide whether its evidence base is sufficient.

The legal hallucination literature is closer in motivation. Dahl et al. (2024) find hallucination rates between 58% and 88% when models are asked verifiable questions about federal court cases, and report that models cannot always predict when they are hallucinating. Magesh et al. (2025), in a preregistered evaluation, find that commercial legal research tools hallucinate less than a general-purpose chatbot but substantially more than their vendors claim. Both measure unsupported output about the law. We measure unsupported output about a regulated party, where the support that is missing is a record rather than a precedent.

Immigration-specific datasets. ImmigrationQA provides 17,058 question-answer pairs generated from United States immigration sources including the USCIS Policy Manual and 8 CFR (Shportun, 2026). ImmigrationReason structures 12,375 non-precedent decisions of the USCIS Administrative Appeals Office, recording for each the applicable legal framework, per-criterion evidence-sufficiency findings under a five-category label, and verbatim adjudicator criticism, including nearly 9,000 instances of legal error identified by the Office (Afsharrad and Mousavi, 2026). ImmigrationReason is nearer to our task than a policy-QA dataset, since it labels whether evidence sufficed for a criterion. Its labels, however, are extracted from the text of adjudications: the sufficiency judgement is the adjudicator's, made about a petition record that the dataset does not contain. Both datasets are text-side. Employer-side obligations, where the per-worker civil penalties and the audit exposure sit, are absent. We are not aware of a public dataset of employer immigration-compliance records with labelled obligation outcomes, and the reason is straightforward: real ones cannot be published.

Compliance benchmarks for AI systems. Automated compliance checking predates language models. In construction, regulatory text is converted into machine-checkable rules by semantic information extraction (Zhang and El-Gohary, 2016); in data protection, privacy policies are checked for completeness against GDPR provisions (Amaral et al., 2022) and questioned directly, with expert analysis of which questions a policy can answer at all (Ravichander et al., 2019). Recent LLM work divides into two directions. In the first, the AI system is the regulated subject: COMPL-AI translates the EU AI Act into technical requirements and benchmarks twelve LLMs against them (Guldimann et al., 2024). The direction of assessment is inverted relative to ours. In the second, the AI system is an assessor of someone else. HSE-Bench poses over 1,000 IRAC-structured health, safety and environment compliance questions (Wang et al., 2025). Principle-Bench evaluates LLM judges of financial promotions against two regulatory principles under a pre-registered rubric (Sarkar, 2026). ReguSim's monitoring benchmark asks a model to judge a marked record within a

window of trading records, and finds that simple structured baselines match or exceed prompt-only LLMs (Luo et al., 2026). These share our assessor-side orientation, and the last shares our record-level input. None of them gives insufficiency of the record its own correct label.

Two recent benchmarks do, and they are the nearest neighbours of this work. DEMM-Bench asks whether agent-runtime records (traces, ledgers, policy logs and the like) are sufficient to reconstruct properties of a specific decision, labels cases with a construction oracle, and reports an overclaim rate as its primary diagnostic (Solozobov, 2026). Its "container fallacy", treating the presence of an evidence container as sufficiency, is close to what we call adjacent-field substitution, where a populated column beside the missing evidence is read as the evidence. Partial Evidence Bench constructs deterministic synthetic scenarios, including a compliance-audit family, in which material evidence lies outside an agent's authorisation boundary, and scores *unsafe completeness*: an answer that appears complete when it is not (Tallam, 2026). It reports that explicit fail-and-report behaviour removes unsafe completeness without collapsing into trivial abstention. That is, in substance, our argument that the false-green rate and the abstention rate must be read together. We differ in three respects. Our subject is a third-party employer and our output a regulatory verdict per (obligation, entity) pair, not a governance reconstruction or a retrieval answer; our insufficiency is a property of what the employer's exports contain, not of what an agent is permitted to see; and the assessment is legal, against cited regulatory obligations. We do not claim the directional, construction-labelled framing as new; both works reach it independently.

Abstention and selective prediction. This is the closest established literature and deserves the most precision, because a reviewer will reasonably ask what is new.

The reject option goes back to Chow (1970), who characterised the optimal tradeoff between recognition error and rejection. Selective classification formalises it as a risk-coverage trade-off (El-Yaniv and Wiener, 2010), and Geifman and El-Yaniv (2017) construct selective deep classifiers that reject as needed to meet a user-set risk level. In this framing coverage is a good to be maximised subject to a risk budget, and abstention is a *cost* incurred to avoid error.

For language models, abstention is evaluated mostly on questions. SQuAD 2.0 adds unanswerable questions to reading comprehension (Rajpurkar et al., 2018); SelfAware pairs unanswerable questions with answerable counterparts to measure self-knowledge (Yin et al., 2023); AbstentionBench aggregates 20 datasets of unknown-answer, underspecified, false-premise and similar questions, and finds that scaling helps little and that reasoning fine-tuning degrades abstention by 24% on average (Kirichenko et al., 2025). Wen et al. (2025) survey the area and organise it by query, model and human values. On the model side, Kadavath et al. (2022) show that models can be trained to predict whether they know an answer. The work nearest ours in this line is Joren et al. (2024), who define whether a retrieved context is *sufficient* to answer a query and find that strong models often answer incorrectly rather than abstain when it is not. That is a statement about the evidence base rather than the query, as ours is, though in open-domain retrieval rather than over an organisation's records. Hallucination benchmarks that measure falsehood against world knowledge (Lin et al., 2022) or the share of atomic claims supported by a knowledge source (Min et al., 2023) address a related failure without an abstention label.

Three things differ here.

First, abstention is not a cost. For 48.8% of pairs in our released set it is the *correct output*, and it is actionable: `needs_evidence` names a document the firm must produce. A system that trades abstention for coverage is not moving along a tradeoff curve, it is producing wrong answers that happen to look like answers.

Second, insufficiency is a property of the evidence base rather than the query, so it is determinable **by construction**. The generator withholds a required export and the label follows mechanically. Most abstention benchmarks annotate insufficiency, and annotation of legal sufficiency is exactly the judgement call that makes such datasets small and contestable. Joren et al. (2024) and the two benchmarks above are partial exceptions: the first classifies sufficiency with an autorater, the latter two fix it by construction.

Third, the error is directional in a way the risk-coverage framing does not capture. Answering `fail` on an undecidable pair is wrong but triggers review. Answering `pass` is wrong and *suppresses* review. Aggregating both into a single risk term hides the only distinction that matters operationally, which is why we report the false-green rate as a separate quantity rather than as one minus abstention recall.

Reasoning over structured and tabular data. Table-based fact verification asks whether a statement is entailed or refuted by a table (Chen et al., 2020), and text-to-SQL asks a system to answer questions over relational databases (Yu et al., 2018). Both originally assumed that every item could be decided. The three-way label of FEVER, with Supported, Refuted and NotEnoughInfo assigned by annotators (Thorne et al., 2018), is the nearest label structure to `pass`, `fail` and `needs_evidence`; FEVEROUS extends it to evidence in table cells (Aly et al., 2021). Unanswerable questions over structured data appear in EHRSQL, a text-to-SQL benchmark over electronic health records (Lee et al., 2022), in work categorising ambiguous and unanswerable text-to-SQL questions (Wang et al., 2023), and in TrustSQL, which scores abstention on infeasible questions with penalties for incorrect output (Lee et al., 2024). ClaimDB is the closest: claims verified against real multi-table databases, with a not-enough-info class, and a finding that both open and closed models struggle to abstain when there is no evidence to decide (Theologitis et al., 2026). Our setting shares the structured input and the insufficiency class. It differs in that the item to be decided is a regulatory obligation attached to an entity rather than a natural-language claim or question, in that labels are fixed by a generator rather than annotated or derived from real databases, and in that the two wrong answers on an undecidable item are scored asymmetrically.

Output format and schema constraints. Our schema ablation removes the abstention verdict from the output format. We run it in three forms (Sections 7 and 8.4): a dashboard schema that also rewrites the verdict definitions, a condition that removes only `needs_evidence`, and a condition that removes every non-committal verdict while keeping the evidence-anchored definitions. Related work shows that output constraints change model answers. Tam et al. (2024) find that restricting generation to structured formats degrades reasoning, more so as constraints tighten. Röttger et al. (2024) show that models forced into a multiple-choice format give substantively different answers from those they give unforced, and that the answers depend on how they are forced. Sclar et al. (2024) report swings of up to 76 accuracy points from meaning-preserving changes to prompt format, which is one reason our prompt is frozen and identical across conditions. Closest in argument is Kalai et al. (2025), who attribute persistent hallucination partly to grading that gives no credit for expressing uncertainty, so that guessing improves scores, and propose changing how existing benchmarks are scored. Our ablation is an instance of the same incentive located in a product interface rather than a leaderboard: a schema with no state for "could not tell" forces a guess, and some guesses are green. We are not aware of prior work that measures the effect of removing the abstention label, holding the rest of the prompt fixed, on a directional error rate.

Aggregate consistency. We searched for prior work measuring whether a system's reported headline counts agree with its own per-item outputs and did not find any. We include it as a metric because we have observed the failure in deployed compliance dashboards, where a card and its drilldown disagree; we flag it as the least-established axis in the suite.

Positioning, stated plainly. The contribution is not a new abstention algorithm and not a theoretical result. Nor is it the first benchmark to fix evidence insufficiency by construction or to score the overclaiming direction separately: DEMM-Bench and Partial Evidence Bench do both, in agent governance and authorisation-limited retrieval. What this benchmark adds is a task in which correct abstention is a large share of the pairs by design and the assessed party is a third-party employer described by structured exports; an instrument for the directional error in that setting, together with an ablation that isolates how much of it is produced by the output schema; and a releasable artifact in a domain where releasability is the binding constraint. Ground truth by construction has precedent in synthetic task suites such as bAbI (Weston et al., 2015); here it also means fresh firms can be drawn at any seed, which addresses the per-benchmark contamination concern raised by Sainz et al. (2023). Constructed labels are not error-free, and test-set label errors can change model rankings (Northcutt et al., 2021), which is why we audit the withheld labels (Section 6). The release follows datasheet practice (Gebru et al., 2021), and our reporting requirement, that the false-green rate always appear beside violation recall, is in the spirit of model cards (Mitchell et al., 2019). Section 11 states what this does not

establish.

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3. Background: the four states

The four-state vocabulary comes from the source engine (Section 5), where it was adopted for a practical reason. A system that reports on legal obligations without being counsel should not state a conclusion it has no basis for, and unauthorised-practice rules (for example ABA Model Rule of Professional Conduct 5.5 and the state statutes it reflects) make the line between reporting records and giving legal conclusions consequential. A separate state for "not assessable from the evidence provided" keeps that line visible.

`pass`**.** A rule was evaluated against present evidence and the evidence satisfies it. The claim must be time-bound, because a pass is only as current as the last data import.

`fail`**.** A rule was evaluated against present evidence and the evidence violates it. The claim must be framed as a potential issue requiring review rather than as a legal conclusion, because characterising conduct as unlawful is an act of legal judgement.

`needs_evidence`**.** The rule could not be evaluated because a required input is absent. This is the state that most often collapses in practice, and the collapse has a direction: a file that was never provided is reported as a file that is in order. The two are opposite. A not-provided document is an **open gap** that counts against audit readiness.

`unmonitored`**.** The obligation is not assessable from this class of source data at all. An employer export cannot evidence whether an adverse action was taken against a worker while a contested tentative nonconfirmation was pending, because that fact lives in conduct and not in a record the employer exports. Such an obligation must remain visible as unmonitored. Silent omission is itself a false green. In the data evaluated in this paper no ground-truth label is `unmonitored`, so the state is defined and permitted but not exercised as a target; a model's `unmonitored` answer on a withheld pair is scored as wrong but is not a false green.

The distinction between `needs_evidence` and `unmonitored` matters operationally: the first is actionable by the firm (provide the document), the second is not (no document exists to provide), and conflating them either sends firms on impossible errands or hides real gaps.

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4. The task

An instance is a firm $F = (S, E, R, a)$ where S is a set of source exports, E a set of entities, R a rulebook, and a a pinned evaluation date.

An **entity** $e in E$ has a type: `worker`, `lca`, `worksite`, `wage_cohort`, or `employer`. Entity type matters because obligations attach at different granularities: an I-9 timeliness duty attaches to a worker, a wage-floor duty to a filing, a prevailing-wage comparison to a cohort defined by occupation, level, and location, and a site-visit readiness duty to the employer as a whole.

An **obligation** $r in R$ carries a regulatory citation, the entity type it attaches to, and a set $mathrm{req}(r)$ of source fields that must be present for the obligation to be decidable. Two conventions qualify it. Reference knowledge (calendars, federal holidays, regulatory thresholds) is supplied by the system rather than required from S, and fields are matched to $mathrm{req}(r)$ by meaning rather than by name. With those conventions, `needs_evidence` is defined by the data rather than by an annotator's sense of sufficiency: if an item of $mathrm{req}(r)$ is absent from S for entity e, the pair is not decidable, and any verdict other than `needs_evidence` is wrong regardless of whether the underlying situation happens to be compliant. Applying the meaning-match is itself a judgement, which is why the anchor labels were audited (Section 6.1).

A **pair** is (r, e) where e's type matches r's attachment. The prediction target is a verdict $v in {texttt{pass}, texttt{fail}, texttt{needs_evidence}, texttt{unmonitored}}$, optionally with evidence pointers and a citation.

All date-relative obligations evaluate against a rather than against the current date, so planted expiry windows remain valid indefinitely. The anchor firm pins a = 2026-06-29.

Scoring rule. A pair with no prediction is counted as an error rather than skipped, so declining to answer lowers violation recall and every accuracy-type score. It does not raise the false-green rate, because an unanswered pair cannot be a false green; every results table therefore also reports coverage and the false-green rate among answered pairs (Section 7).

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5. Benchmark construction

5.1 The rulebook

The rulebook contains 194 obligations across ten regimes: I-9 and E-Verify (28), Labor Condition Application (23), Public Access File (15), wage and prevailing wage (21), H-1B status (16), I-129 supplement (13), L-1 (21), F-1 OPT and STEM OPT (15), site-visit readiness (14), and cross-cutting deadlines and penalties (28). The source manifest lists 196 entries; two rules appear deliberately in two regimes, once as a field check and once as a deadline clock, and are counted once here under I-9.

Each obligation carries the check it performs, its threshold, its regulatory citation, and a description of the information needed to decide it. The citations had previously been through a verification pass by the source engine's maintainers (the author's company), in which incorrect citations and thresholds were corrected against primary sources; it was not an independent legal review. The rulebook is built programmatically from the source engine's machine-readable registry of that manifest rather than retyped, so every citation is traceable to a source document rather than to our recollection.

The anchor firm's expectation set also refers to 17 rules by the engine's internal identifiers rather than manifest identifiers. Fourteen map to the manifest rule stating the same duty through the engine's own published alias table. Three are engine roll-ups whose definitions are assembled from the manifest rules they aggregate and carry those rules' citations. No obligation text in the benchmark is written from scratch.

5.2 The anchor firm

The anchor firm is a synthetic employer of twelve workers, nine on H-1B status and three United States citizens, operating from two worksites, with eleven source exports totalling 93 data rows: HR roster, case-management export, certified LCAs, I-9 records, E-Verify cases, payroll register, RFE and NOID cases, cap cases, green-card cases, PERM recruitment tracker, and benefits census.

Most workers carry one planted regulatory mechanism, so that a missed finding usually localises to a mechanism. Representative plants:

Mechanism	Planted signal
I-9 Section 2 timeliness	Section 2 completed 8 business days after start, against a 3-day limit
Reverification deadline	Employment authorisation document expired 14 days before the pinned date, with no reverification recorded
E-Verify TNC window	Notice delivered, no employee decision recorded, well past the response window
LCA expiration	Certified period ending 4 days after the pinned date
Required wage, greater-of	Salary equals the certified rate but falls below the prevailing wage

Mechanism	Planted signal
Material change, occupation	Filing carries an individual-contributor occupation code against a management title
Material change, worksite	Roster location differs from the filing's stated area of intended employment
Petition expiry	Validity ending 21 days after the pinned date

What the anchor labels are, and their known problems. The anchor firm's labels were written as the acceptance gate for a compliance engine, not generated, so they are not "fixed by construction" in the sense the generated firms are. Their `fail` means "requires action" in that engine. A review of the 20 violation labels while preparing this paper found the following, all disclosed rather than silently corrected, because the labels predate every result:

- *Duplicates.* Four facts each carry two violation labels under different rules: Mei Tanaka's wage shortfall, Sofia Reyes's worksite, Daniel Cho's E-Verify case, and Hiroshi Yamamoto's petition expiry. The 20 labels describe 16 distinct facts.
- *Not violations on the records.* Five "actual-wage memo missing" labels have no evidence in any export, and five are deadlines that had not passed on the evaluation date. The strict view (Section 6) handles both.
- *Contestable plants.* Some plants ignore exceptions a practitioner would check. A roster location outside the LCA's area may be a permitted short-term placement (20 C.F.R. 655.735). A management title against an individual-contributor occupation code is not by itself a material change, since the test is a change in duties (8 C.F.R. 214.2(h)(2)(i)(E)). Helena Voss's reverification label has no reverification data in the export. One certified LCA states an offered wage below its prevailing wage, which the Department of Labor would not certify.

We keep the engine's labels as the headline view because they existed before this study. The strict view, the duplicate count and the generated firms are the checks on them.

The firm's expectation set contains no `pass` labels. Some workers' records conform, but the engine's expectations name only violations and evidence requests, so the anchor firm cannot measure false alarms; the generated firms can (Section 5.3).

Its expectation set contains 221 pairs: 20 planted violations and 201 evidence requests. The 201 arise because most obligations in this domain require evidence an employer export does not contain. A certified LCA does not evidence that the required notice was posted; a roster does not evidence which document a worker presented. The honest output for such a pair is an evidence request. The ten-to-one ratio reflects which obligations the source engine can assess from exports, not a measured property of real employers; we treat the large abstention share as a design choice that mirrors that engine's coverage.

5.3 The generator

The anchor firm is one employer, too small to support an interval on any system's error rate and lacking conforming controls. The generator addresses both, over a deliberately narrow set of rules.

Its central property is that **labels precede data**. For each applicable (obligation, entity) pair the generator first draws a disposition and then writes records that decide it:

- `planted_violation`: the records show the breach;
- `planted_control`: the records show conformity, with a configured share placed exactly on the legal boundary (near-miss controls) to test whether a system flags on proximity rather than on the rule;
- `withheld_evidence`: the export that would carry the required information is not provided to the firm at all.

The generator emits a pair only for the four rule families whose every listed input it renders: I-9 Section 1 timeliness (signed on or before the first day of employment, not before offer acceptance), I-9 Section 2 timeliness (within three business days, weekends and federal holidays with their observed dates excluded), reverification on or before the expiry of temporary work authorisation, and the LCA required wage (wages paid at least the greater of the prevailing wage and the employer's actual wage for similar workers, from a semimonthly payroll register and an actual-wage table). Each rule is applied only to entities it governs. Near-miss controls sit on the boundary: Section 1 on the first day, Section 2 on business day three, Supplement B on the expiry date, pay exactly equal to the required wage.

Evidence is withheld a whole export at a time, never by blanking one worker's cell. A blank date for a past deadline reads to a practitioner as "not done", which is a violation, not missing evidence. A firm either provides its I-9 export, and every I-9 pair is decidable, or does not, and every I-9 pair is `needs_evidence`.

An earlier generator failed this standard, and we replaced it. A first version emitted a label for every (obligation, entity) pair in the 194-obligation rulebook but rendered deciding data for only four rule families, and applied rules to entities they cannot govern (L-1 and OPT rules to H-1B workers). A blind consistency check caught this before any language-model result on generated firms was reported. An AI agent (Claude) given only the records and the rule text for 60 stratified pairs, and not the labels, agreed with its labels on 8. Every figure from that version, including a released-set size of 58,961 pairs and the trivial baselines computed on it in an earlier draft, is withdrawn; the artifacts are archived in `docs/study2_v1_invalid/`. The same check on the current generator, on 60 pairs from seeds not otherwise used, agreed on all 60 (15 per stratum). The test suite now checks every rendered record against its label across 40 seeds.

The cost of this design is breadth. The generated firms test four rule families thoroughly, including the boundary cases, while the anchor firm's breadth (221 pairs across eight regimes) comes from hand authoring. Generation is deterministic in the seed, which the test suite asserts.

5.4 Released scale

	Value
Firms	31 (1 anchor + 30 generated at seeds 1-30)
Pairs	1,481
Withheld-evidence pairs	723 (48.8%)
Planted violations	238
Obligations in the rulebook	194 (the generator renders four rule families)

Additional firms are available at any seed.

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6. Metrics

Let T be ground truth and P predictions over pairs.

Violation recall. Of pairs labelled `fail`, the share predicted `fail`. A missed violation is the failure the domain treats as unforgivable, so this is reported first.

Violation precision and **false-alarm rate.** Precision over `fail` predictions, and separately the share of conforming controls (label `pass`) on which a system returned anything other than `pass`: `fail`, `needs_evidence` and `unmonitored` all count as an alarm, since each sends a compliant item to review. Near-miss controls, placed exactly on the legal boundary, are reported separately in Study 2.

False-green rate. The headline metric:

mathrm{FGR} = frac{|{p : T(p) = texttt{needs_evidence} wedge P(p) = texttt{pass}}|}{|{p : T(p) = texttt{needs_evidence}}|}

The share of undecidable pairs a system reports as satisfied. Note what it is not: it is not one minus abstention recall. A system that answers `fail` on an undecidable pair is wrong, but it is wrong in a direction that triggers review. Only `pass` suppresses review, so only `pass` counts here.

Abstention precision and recall, treating `needs_evidence` as a class. Precision matters because a system can drive the false-green rate to zero by abstaining everywhere, which the baselines demonstrate.

Two further axes are defined in the harness for systems that expose them, and are **not measured in this paper's studies**. *Evidence-pointer accuracy* is the share of correct verdicts whose cited evidence intersects the fields ground truth records as relevant. *Consistency drift* is the disagreement between the counts a dashboard displays and its own per-pair verdicts. Neither applies to the language models evaluated here; we list them so that system builders can report them.

Reporting requirement. The false-green rate must be reported alongside violation recall. Neither is meaningful alone, and Section 8 shows why.

6.1 Label views of the anchor firm

Two label views, fixed before per-pair results were read. The anchor firm's labels were written as the acceptance gate for a compliance engine, where `fail` means "requires action". This benchmark's `fail` means "the records show a violation". Ten of the 20 planted violations sit between those meanings, so we score under two views, fixed in `nofalsegreen/labelviews.py` on 2026-09-24 with each affected pair named by identifier. The commit came six minutes after a first, later-discarded attempt at the main run had printed preliminary aggregate rates to its log. No per-pair result had been read, but we cannot exclude that those printed aggregates influenced the decision, and say so.

- **Engine view, the headline.** The labels as authored. It is the headline because it existed before this benchmark did, so nothing decided here can have tuned it.
- **Strict view.** Five "actual-wage memo missing" violations become `needs_evidence`: no export records the memo as absent, the evidence is simply not there, which is this benchmark's own definition of an evidence request. Five deadlines that had not yet passed on the evaluation date, an LCA and a petition near expiry, an open extension window, and an RFE and a NOID not yet due, are removed from scoring and reported separately as deadline alerts, listing the verdict each model gave. The remaining ten violations are breaches that had already occurred.

The strict view's violation recall has a denominator of ten, so a single pair moves it by 0.1. It is always reported beside the engine view, never alone. A model that answers `needs_evidence` on the unevidenced memos loses recall under the engine view and none under the strict view; the test suite asserts exactly that, so the difference between the two views on those pairs is a direct measure of how often a model declined to call a missing document a violation.

A third view, from a blind audit of the withheld labels. While reading early results we found a withheld label the exports contradict. `i9.section2.employer_signature` is labelled `needs_evidence` because the source engine never maps the column, yet `section2_signed_date` and `section2_examiner_name` are populated for every employee, and the prompt tells models to match columns by meaning. Because this was noticed in model output, the correction is constrained so that it cannot be steered by that output. The protocol (`docs/LABEL_AUDIT_PROTOCOL.md`) was committed before the audit ran. It covers all 120 withheld pairs, not the ones models got wrong. The auditor is a separate agent given only each pair's obligation text and the exports its batch was shown, with no model reply, score, or engine note. A label is overturned only if every required item of "information needed" is present for that entity; ambiguity keeps the label.

The audit overturned 2 of 120 labels, the Section 1 and Section 2 I-9 signatures, both on signed-date columns populated for all twelve employees. The auditor flagged the Section 1 call as the weaker: there is no separate signature flag, and a

stricter reader could hold that a date is not a signature. In the **audited view** those pairs are removed from scoring, not relabelled, since an auditor's verdict is not a label fixed by construction. Removing pairs can only remove false greens, so every table reporting this view states how many pairs it scores, and the engine view remains the headline. Every audit decision is published with its reasoning (`docs/label_audit.json`).

7. Baselines

Trivial baselines (run). `always_pass` answers `pass` everywhere and is maximally unsafe. `always_abstain` answers `needs_evidence` everywhere: safe but useless. `always_fail` flags everything. `oracle` reads the label basis and decides exactly when evidence is present; it is not a competitive system but an upper bound establishing that the task is solvable.

Language-model baselines. The harness was tested against mock models with known failure signatures before any real call was made. The protocol is frozen in code (`baselines/prompts.py`, `PROMPT_VERSION` 1.1.0) so that the reported result cannot be a post-hoc selection of prompt:

- *Prompt.* Per firm, the source exports verbatim as CSV, the obligations needed by the batch with their check, threshold, citation and the information needed to decide them, and the pairs to decide. Two reading rules apply identically in both conditions: export columns are matched to obligations by meaning, and reference knowledge such as federal holidays and regulatory thresholds is supplied by the model rather than required from the exports. Verdicts are returned one per line as `pair_id`, verdict, evidence field. The evaluation date is pinned in the prompt with an explicit instruction not to use the real current date, since every deadline rule depends on it.
- *Output schema ablation.* `abstain_offered` permits four verdicts, defines `pass` as "the required evidence is present in the exports and satisfies the obligation", and adds a paragraph stressing the `pass`-versus-`needs_evidence` distinction. `abstain_withheld` permits only `pass` and `fail` and is a dashboard-style schema in which every cell must resolve. It changes four things at once: it removes `needs_evidence`, removes `unmonitored`, redefines `pass` as "the obligation is satisfied", and drops the paragraph. The comparison therefore measures the effect of moving from an evidence-aware schema to a dashboard schema, not of one label. To isolate the label, the Vertex block (Section 8.4) adds `abstain_removed_only` (the offered prompt word for word, minus the `needs_evidence` verdict and the sentences that define it) and `abstain_closed` (only `pass` and `fail`, with their evidence-anchored definitions and the warning kept). We count **off-schema** replies, where a model returns a verdict the schema does not permit.
- *Decoding.* Every request sets temperature 0 and an output cap of 32,000 tokens; for models that reason before answering, the reasoning tokens count toward the cap. Some free-tier hosts may not honour temperature 0 for reasoning models, which is one reason the Vertex block repeats every run three times.
- *Models.* Study 1 ran six free API models from four providers (Section 7.2). The Vertex block adds three Google models: Gemini 3.1 Pro, Gemini 3.8 Flash and Gemma 4 26B. Results speak to these nine models only.
- *Aggregation.* Scored per firm, then averaged across firms with standard deviation. Pairs within a firm share source records and are not independent.

Parsing policy, which is a scoring decision. A reply line naming a pair but none of the four verdict words yields **no prediction**, and the metric module counts the pair as an error. It is never coerced. Coercing to `needs_evidence` would convert a model's formatting failures into abstentions and flatter its false-green rate; coercing to `pass` would do the reverse. Either would make the headline number a function of our parser. We report the unparseable rate separately so a reader can see how much of a score is formatting. A line carrying a recognised verdict that the schema did not permit (for example `needs_evidence` under `abstain_withheld`) is parsed as given and counted as off-schema.

Coverage, because an unanswered pair cannot be a false green. A pair with no prediction scores as a miss on violation recall, which is conservative. On the false-green rate it is the opposite: the pair stays in the denominator and

can never be counted as `pass`, so a model that answers less looks safer, and one that answers nothing scores a perfect 0.000. The headline rate keeps the declared definition, but every results table also reports how many pairs each model answered and the false-green rate among withheld pairs it actually answered. A gap between the two columns is a coverage effect, not safety.

Delivery failures are not answers. Free endpoints sometimes return an empty reply, end a reply with an upstream error, or cut it off below our output cap. None of these is the model's answer, so the batch is re-requested, at most three attempts in all. Three such failures in a row, across batches, means the host is failing rather than the batch: the run stops, keeps its complete batches, and resumes after the next quota reset. A run is never recorded as finished while a batch is missing for this reason, because its missing pairs would score as model errors when the cause was delivery. Two outcomes are the model's own and are kept as given: a reply that ends normally but contains no usable verdict lines (counted as unanswered), and a reply that runs to our 32,000-token cap on all three attempts (the cap is close to, and below, the free models' own output ceilings, so the model is generating without end; the batch is recorded as unanswered). Every attempt, with its token count and the provider's stop reason, is in the sweep log.

Two departures from this rule occurred, and both are disclosed here. First, Nemotron 3.5 Lightning's second run never finished normally: its host repeatedly cut replies off at a ten-minute limit. Under a cut-off rule committed before the final quota window, once every other run had finished, the run was given one pass that tried each remaining batch up to three times and was then recorded with what it had answered: 68 of 140 pairs. Second, during that pass one reply ended with an upstream error *after* delivering all twelve verdict lines. The harness accepted it because the reply was not empty, contrary to the rule above. We keep it, since its answer is complete, and the harness has since been corrected to re-request such replies. It is the only reply in the study that ended without a normal stop.

Cost. Study 1 and Study 2 used free API tiers only. The Vertex block was paid from the author's startup cloud credits; token counts per run are in `docs/vertex/runs/`.

7.1 Protocol revision after a pilot

A pilot under prompt v1.0 exposed three defects in our own harness, not in the models: obligations sent without their definitions, a mangled "information needed" field, and a definition of `needs_evidence` that a literal reader could apply to every pair. We fixed them in prompt v1.1.0, archived the pilot, and report no pilot number as a result. The main runs then exposed an output cap that cut off reasoning models mid-reply; incomplete replies were purged and re-requested under a larger cap. Both revisions, with the numbers that motivated them, are in Appendix C.

7.2 Running on free API tiers

We report results obtained entirely on providers' free tiers, which constrains the experiment in two ways that must be stated because both affect the measurement.

Source scoping. Free tiers cap tokens per minute, and the cap is checked against the largest single call, not the average. Sending all eleven exports on every call exceeds several free per-minute token quotas, so we send only the exports whose regime an obligation can be evaluated against, with the HR roster always included because every other source joins to it. Every run in this paper uses batches of 12 pairs.

Scoping is a change to the prompt, so we treat it as part of the protocol rather than an implementation detail. Two safeguards keep it from manufacturing abstentions. First, the prompt names the exports that were withheld and instructs the model not to infer that an absent field exists in an export it cannot see; a model that abstained merely because it suspected hidden data would otherwise inflate the apparent abstention rate. Second, an obligation whose regime is unclassified falls back to all sources, so the mapping can only narrow prompts for regimes it knows about and can never hide data from an obligation it has not classified. We did not run models with scoping off, so we cannot rule out that scoping changes model behaviour; it is applied identically in every condition, so it does not bias the comparison between them.

Model coverage. Free tiers offer a narrower and generally smaller set of models than a funded study would use. The Vertex block adds one frontier model, but no model from other frontier providers was evaluated (Section 11).

Model set. Six free models were declared. Two (the Nex models) were withdrawn from the free tier mid-study; they were replaced by Qwen 3.8 27B and, after Gemma 4 31B never passed a screening call, dots.3 Note Preview, under rules committed before any replacement had answered a batch. Appendix C gives the full sequence and its commit record.

The declared evaluation subset. The reason for a subset is the free-tier call budget: the full anchor firm at batch 12 costs 19 calls per model per condition against a ceiling of 50 calls a day. We evaluate a **declared stratified subset** of 140 pairs, fixed by a seed in `nofalsegreen/subset.py` and frozen to a file before any model was scored. It contains **every planted violation** (so violation recall is over the complete violation set) and 120 of the 201 withheld pairs, sampled in proportion to each regime with at least one pair per regime present. The subset's smaller denominator widens every interval; we report its size beside every result and never present a subset rate as a full-firm rate.

A baseline we deliberately exclude. The anchor firm's expectation set originated as the acceptance gate for a commercial compliance engine developed by the author's company (Section 12). That engine would score near ceiling on this firm for a reason that has nothing to do with capability: the expectations and the engine were developed against each other. Reporting it as a baseline would be misleading, and we do not.

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8. Results

8.1 Trivial baselines

Table 1 — trivial baselines. Across 31 firms (the anchor and 30 generated), aggregated per firm then averaged over the firms in which each rate is defined: a generated firm may withhold every export, or none, so not every rate exists in every firm. Reproduced by `python3 scripts/sweep.py --firms 30`.

System	False green	Violation recall	Precision	False alarm	Abstention F1
`always_pass`	1.000	0.000	—	0.000	0.000
`always_abstain`	0.000	0.000	—	1.000	0.717
`always_fail`	0.000	1.000	0.170	1.000	0.000
`oracle`	0.000	1.000	1.000	0.000	1.000

A dash marks a rate undefined for that system: `always_pass` and `always_abstain` never predict a violation, so their precision has no denominator.

Three readings.

The false-green rate is not redundant. `always_pass` attains a false-alarm rate of 0.000, the best possible score on compliant cases. Under any metric that rewards not crying wolf, it is indistinguishable from a careful system. It is in fact the worst possible system for deployment, reporting all 723 undecidable pairs as satisfied. Only the false-green rate separates them. On the anchor firm alone the separation is starker: `always_pass` reports all 201 unassessed obligations as satisfied while raising zero false alarms.

The false-green rate is also not sufficient. `always_abstain` attains 0.000 on it while detecting no violations, and reaches 0.717 abstention F1. On the anchor firm, whose expectation set declares no control pairs, it reaches 0.953 abstention F1 while being useless. Abstention scores are therefore gameable in isolation, which is why the suite must be reported jointly.

The task is solvable. The oracle reaches 1.000 on every axis. A system scoring poorly is exhibiting a property of itself, not of an ill-posed benchmark.

8.2 Language-model results

Six free models were evaluated on the declared 140-pair subset of the anchor firm under prompt v1.1.0, each under both output schemas: Nemotron 3 Ultra 550B, Nemotron 3 Super 120B and Nemotron 3.5 Lightning (NVIDIA), Laguna XS 2.1 (Poolside), Qwen 3.8 27B (Alibaba), and dots.3 Note Preview (dots-studio), the last two replacing the withdrawn Nex models under the rule in Section 7.2. Gemma 4 31B, the other declared replacement, never passed the screen and yielded its slot to the fallback. The tables are generated from the retained replies (`scripts/fill_results.py`); the coverage and answered-only columns were added after the first results, when the coverage effect described in Section 7 was found.

Uncertainty, and what it does and does not allow. Bracketed figures are 95% percentile intervals from a bootstrap that resamples *obligations* (2,000 draws, fixed seed): the pairs under one obligation share a rule text and records, so resampling pairs would understate the interval. The intervals are wide. At 140 pairs most differences *between models* within a condition are not resolved, so we do not rank the models, and tables list them alphabetically. The comparison the study is built for is *within* a model across the two schemas. We bootstrap the paired difference on the withheld pairs both runs answered, and correct the six per-model tests for multiplicity (Holm). All six remain significant at 0.05 (Table 3). Every Study 1 run was made once; the Vertex block (Section 8.4) measures run-to-run variation, and Study 2 (Section 8.3) adds six generated firms.

Table 2 — main result, `abstain_offered`. One row per model, scored on the declared 140-pair subset of the anchor firm. The protocol averages across firms with a standard deviation; with one firm evaluated, each cell is that firm's value. The unparseable and answered columns guard the rest.

Model	False green	FG, answered only	Recall	Precision	False alarm	Abst. P	Abst. R	Abst. F1	Unparse.	Answered
`dots-3-note-preview`	**0.100** [0.03, 0.21]	0.100	0.450 [0.20, 0.79]	0.900	0.000	0.906	0.883	0.894	0.000	140/140
`laguna-xs-2.1`	**0.125** [0.05, 0.24]	0.135	0.200 [0.04, 0.45]	0.667	0.000	0.940	0.783	0.855	0.000	124/140
`nemotron-3-super-120b-a12b`	**0.050** [0.01, 0.11]	0.050	0.300 [0.10, 0.61]	0.857	0.000	0.896	0.858	0.877	0.000	140/140
`nemotron-3-ultra-550b-a55b`	**0.067** [0.02, 0.14]	0.067	0.300 [0.10, 0.60]	1.000	0.000	0.903	0.933	0.918	0.000	140/140
`nemotron-3.5-lightning`	**0.042** [0.01, 0.09]	0.059	0.050 [0.00, 0.18]	0.500	0.000	0.951	0.642	0.766	0.000	92/140
`qwen3.8-27b`	**0.142** [0.06, 0.25]	0.142	0.350 [0.14, 0.67]	0.875	0.000	0.894	0.842	0.867	0.000	140/140
`always_pass` (ref)	1.000	1.000	0.000	—	0.000	0.000	0.000	0.000	0.000	140/140
`oracle` (ref)	0.000	0.000	1.000	1.000	0.000	1.000	1.000	1.000	0.000	140/140

The trivial baselines are repeated in-table as reference rows, so a reader sees immediately where each model sits between the unsafe floor and the attainable ceiling. The false-alarm column is 0.000 for every row by construction, since the evaluated subset has no conforming pair (Section 9); it is kept so the table matches its declared shape.

Offered an abstention, models abstain readily and detect little. False-green rates run from 0.042 to 0.142. The lowest belongs to Lightning, which answered only 92 of 140 pairs; among the withheld pairs it actually answered its rate is 0.059. No model is close to `always_pass`, and every model is far from the oracle on violation recall, which runs from 0.050 to 0.450: each model misses between 11 and 19 of the 20 violation labels, counting unanswered pairs as misses. As a rate this is the larger failure, against at most 0.142 of withheld pairs reported as compliant. Models answer `needs_evidence` on violations whose evidence is in the exports. Abstention F1 is between 0.766 and 0.918, so the abstentions mostly land on the right pairs; there are too many of them.

Table 2b — strict label view. Same runs, scored under the strict view (Section 6), with the deadline alerts beside them. Generated from `docs/views_results.json`.

Model	Condition	FGR, engine	FGR, strict	Recall, engine (of 20)	Recall, strict (of 10)	Alerts: fail / pass / other
`dots-3-note-preview`	offered	0.100	0.096	0.45	0.70	2 / 0 / 3
`dots-3-note-preview`	withheld	0.525	0.504	0.80	0.80	3 / 2 / 0
`nemotron-3-super-120b-a12b`	offered	0.050	0.048	0.30	0.40	2 / 0 / 3
`nemotron-3-super-120b-a12b`	withheld	0.175	0.168	0.95	0.90	5 / 0 / 0
`nemotron-3-ultra-550b-a55b`	offered	0.067	0.064	0.30	0.50	1 / 1 / 3
`nemotron-3-ultra-550b-a55b`	withheld	0.217	0.208	0.80	0.90	2 / 3 / 0
`nemotron-3.5-lightning`	offered	0.042	0.040	0.05	0.10	0 / 1 / 4
`nemotron-3.5-lightning`	withheld	0.208	0.200	0.50	0.40	3 / 0 / 2
`laguna-xs-2.1`	offered	0.125	0.120	0.20	0.30	1 / 0 / 4
`laguna-xs-2.1`	withheld	0.433	0.416	0.90	0.80	5 / 0 / 0
`qwen3.8-27b`	offered	0.142	0.136	0.35	0.50	2 / 0 / 3
`qwen3.8-27b`	withheld	0.417	0.400	0.85	0.90	3 / 2 / 0

The strict view moves false-green rates by at most 0.021 and raises recall for every model under `abstain_offered`, because the five memo pairs that most models answered `needs_evidence` stop counting as missed violations. A `pass` on a deadline days away is the one deadline-alert outcome a reviewer would never see. Offered an abstention, models gave it on 2 of 30 alerts. Without the abstention it rose to 7 of 30, three of them from Nemotron Ultra, which answered `pass` on an LCA expiring in four days.

Table 2c — audited label view. Same runs, with the two withheld pairs the blind audit overturned removed from scoring (Section 6). The scored denominator is stated in every row.

Model	Condition	Withheld pairs scored	FGR, engine	FGR, audited	FG, audited, answered only
`dots-3-note-preview`	offered	118 of 120	0.100	0.085	0.085
`dots-3-note-preview`	withheld	118 of 120	0.525	0.517	0.517
`nemotron-3-super-120b-a12b`	offered	118 of 120	0.050	0.034	0.034
`nemotron-3-super-120b-a12b`	withheld	118 of 120	0.175	0.161	0.161

Model	Condition	Withheld pairs scored	FGR, engine	FGR, audited	FG, audited, answered only
`nemotron-3-ultra-550b-a55b`	offered	118 of 120	0.067	0.051	0.051
`nemotron-3-ultra-550b-a55b`	withheld	118 of 120	0.217	0.203	0.203
`nemotron-3.5-lightning`	offered	118 of 120	0.042	0.025	0.036
`nemotron-3.5-lightning`	withheld	118 of 120	0.208	0.212	0.439
`laguna-xs-2.1`	offered	118 of 120	0.125	0.110	0.119
`laguna-xs-2.1`	withheld	118 of 120	0.433	0.424	0.472
`qwen3.8-27b`	offered	118 of 120	0.142	0.127	0.127
`qwen3.8-27b`	withheld	118 of 120	0.417	0.407	0.407

Removing the two I-9 signature pairs lowers the false-green rate by 0.008 to 0.016 in 11 of the 12 runs. In the twelfth, Lightning without the abstention, it rises by 0.004, because neither removed pair was one of that run's false greens. It changes neither the ordering of models nor the direction of any condition effect. The label error the audit found is real but too small to account for the result.

Table 3 — the schema ablation. The paper's most informative experiment. Same models, both conditions, difference reported per model.

Model	Withheld pairs answered in both	FGR offered	FGR withheld	Change [95% CI]	Holm-adjusted p	Off-schema replies (withheld)
`dots-3-note-preview`	120 of 120	0.100	0.525	+0.425 [+0.26, +0.59]	0.003	0 of 140 answered
`nemotron-3-super-120b-a12b`	120 of 120	0.050	0.175	+0.125 [+0.03, +0.24]	0.006	0 of 140 answered
`nemotron-3-ultra-550b-a55b`	120 of 120	0.067	0.217	+0.150 [+0.07, +0.27]	0.003	0 of 140 answered
`nemotron-3.5-lightning`	41 of 120	0.042	0.208	+0.463 [+0.12, +0.73]	0.006	0 of 68 answered
`laguna-xs-2.1`	99 of 120	0.125	0.433	+0.333 [+0.16, +0.51]	0.003	0 of 128 answered
`qwen3.8-27b`	120 of 120	0.142	0.417	+0.275 [+0.13, +0.43]	0.003	0 of 140 answered

FGR columns are over all 120 withheld pairs, as in Table 2. The change, its interval (obligation-cluster bootstrap, 2,000 draws) and the one-sided p-value are computed on the withheld pairs both runs answered; the bootstrap p-value has a floor of 1/2,000.

The off-schema column counts answered pairs where a model returned `needs_evidence` or `unmonitored` although the schema permitted only `pass` and `fail`.

Removing the abstention from a dashboard schema raises false greens in every model. The direction of this effect is close to built in. A schema with only `pass` and `fail` forces every unassessable pair into one of the two, so some false greens are guaranteed, and the withheld-condition false-green rate is the share of forced decisions that went to `pass`. Three things are not determined by the design, and they are the result.

The size of the shift varies widely. The false-green rate rises by between 0.125 (Nemotron Super) and 0.425 (dots.3) over all withheld pairs (Table 3). Five of the six models sent most unassessable pairs to `fail`, an unsupported violation

(32 to 99 of 120); dots.3 sent more to `pass` (63) than to `fail` (57). Nothing in the schema says an unassessable pair should be called compliant rather than violated, and models differ in which way they lean.

No model returned an off-schema verdict among the pairs it answered. Lightning's host returned 72 empty or cut-off replies in this condition. Those are delivery failures, not refusals of the schema, and we cannot tell what the model would have answered.

False greens persist when abstention is offered (Table 2).

This condition changes more than the label (Section 7): it also redefines `pass` and drops a warning paragraph. Its effect is therefore the effect of moving to a dashboard schema. The one-variable condition in Section 8.4 isolates the label.

Table 4 — false-green rate by regime, `abstain_offered`. One row per regulatory regime, one column per model. Each cell is the false-green rate over the regime's withheld pairs, with unanswered pairs counted as not-`pass`, and in parentheses the number of the regime's scored pairs of any label.

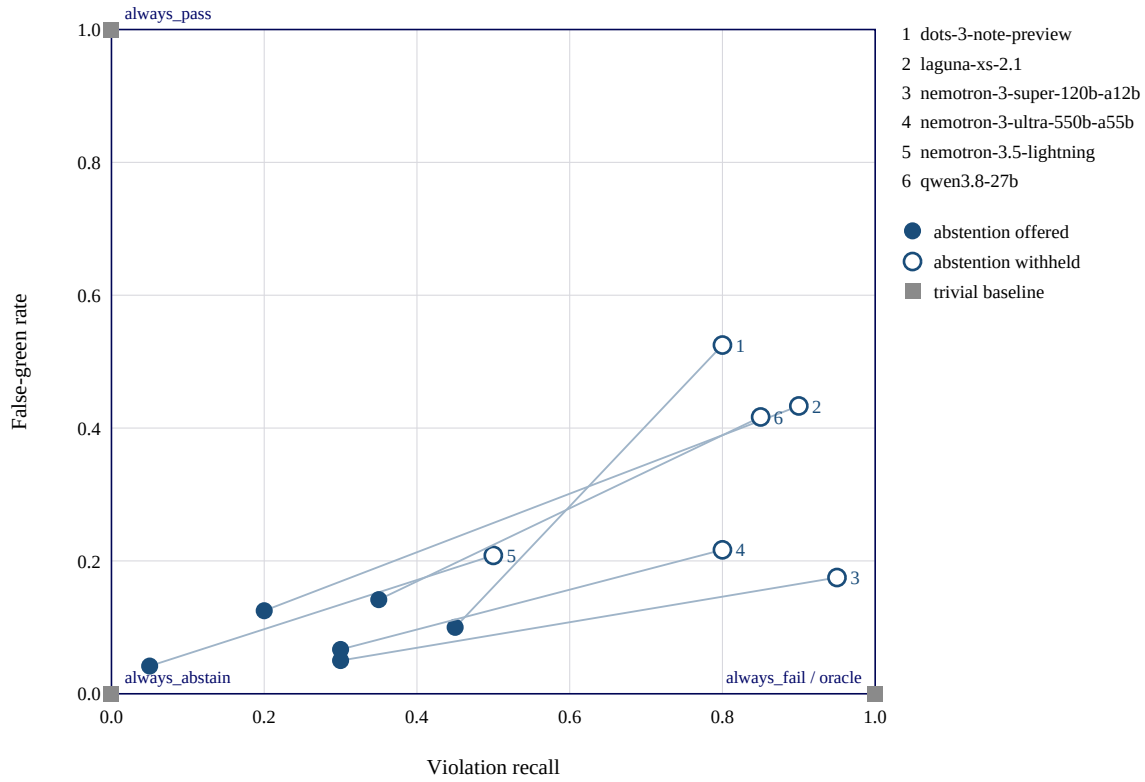
Regime	nemotron-3-ultra-550b-a55b	nemotron-3-super-120b-a12b	nemotron-3.5-lightning	laguna-xs-2.1	qwen3.8-27b	dots-3-note-preview
deadlines-penalties	0.000 (6)	0.000 (6)	0.000 (6)	0.000 (6)	0.250 (6)	0.000 (6)
fdns-sitevisit	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)	0.000 (7)
h1b-status	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)	0.000 (17)
i129-supplement	0.000 (8)	0.000 (8)	0.000 (8)	0.000 (8)	0.125 (8)	0.000 (8)
i9-everify	0.217 (28)	0.174 (28)	0.174 (28)	0.217 (28)	0.304 (28)	0.174 (28)
lca-dol	0.100 (33)	0.067 (33)	0.000 (33)	0.100 (33)	0.100 (33)	0.100 (33)
paf	0.000 (17)	0.000 (17)	0.059 (17)	0.000 (17)	0.000 (17)	0.000 (17)
wage-prevailing	0.000 (24)	0.000 (24)	0.000 (24)	0.389 (24)	0.278 (24)	0.278 (24)

Cells are the false-green rate over the regime's withheld pairs; in parentheses, the regime's scored pairs of any label.

The anchor subset has no conforming pairs, so near-miss controls are measured in Study 2 (Table 6), not here.

False greens are concentrated in a few regimes. Under `abstain_offered`, three regimes, I-9 and E-Verify, prevailing wage, and the LCA rules, hold almost every false green; H-1B status and FDNS site visits have none for any model. The regimes with false greens are, on inspection, ones whose exports carry a populated column *next to* the missing evidence: a document title beside a physical examination, an unitemized deduction total beside an itemised fee check. Section 9 examines this; it is an interpretation formed after the results.

Figure 1 — recall against false-green rate, Study 1. One point per model and condition; filled circles offered the abstention, open circles did not, and a line joins each model's two conditions. Grey squares are the trivial baselines.



Offered an abstention, the models sit low and left: few false greens, low recall. Without it they move up and right together, gaining recall at the cost of false greens. No model reaches the oracle's corner. The figure has a limitation a reader should know: `always_fail` sits at the same point as the oracle, because neither axis penalises calling an unassessable pair a violation. That error is counted in the decomposition of Section 9. A recall-only evaluation would prefer the withheld condition, since it detects more violations; the false-green axis shows what that preference costs.

8.3 Study 2: generated firms

Study 2 runs the same protocol on all 252 pairs of six generated firms (Section 5.3), which carry what the anchor firm lacks: conforming controls, a third of them near-miss cases on the legal boundary. Firms were fixed in `nofalsegreen/study2.py` before any call. The declared free-tier models are still running on the free quota and are not reported here; Study 2 results come from the three Vertex models, three runs each, shown in the Study 2 rows of Table 7 (Section 8.4).

Generated firms are easier, and show the other side of the trade-off. With abstention offered, the three models report 0.021 to 0.071 of withheld pairs as compliant, below their Study 1 rates. Gemini 3.1 Pro and 3.8 Flash detect 0.960 to 0.980 of violations; Gemma 4 detects 0.283. The generated exports are clean, with no populated column next to a missing one, which is consistent with the adjacent-field account of Section 9. The conforming controls expose a cost that Study 1 could not show. With abstention offered, Gemini 3.1 Pro returns something other than `pass` on 0.190 of conforming controls and 0.162 of near-miss controls, mostly by asking for evidence that is present. Without any abstention verdict its false-alarm rate falls to 0.032 while its false-green rate rises from 0.021 to 0.061. The abstention option buys fewer false greens with more review of compliant items.

Excluding the reverification pairs, whose violation labels assume no extension was filed (Section 11), does not lower any Study 2 false-green rate, so that simplification does not drive these results.

8.4 The Vertex block: a frontier model, and repeated runs

Every model in Section 8.2 is a free-tier open-weights model, run once. The Vertex block adds Gemini 3.1 Pro (Google's frontier tier at the time of writing), Gemini 3.8 Flash, and Gemma 4 26B (open weights, Google-hosted), on both studies, and runs every configuration three times (`nofalsegreen/vertex_block.py`). These models were added after the free-tier studies began, and are reported apart for that reason.

It also takes the schema ablation apart. The dashboard condition of Section 8.2 changes four things at once, so the block adds two conditions, each declared before its first call. In *removed-only*, the offered prompt loses only `needs_evidence` and its definition. In *closed*, both non-committal verdicts are removed but `pass` and `fail` keep their evidence-anchored definitions and the warning sentence. The *closed* condition was added after the first *removed-only* results were seen, and is reported as such.

Table 7 — Vertex block. Mean of three runs, with the minimum and maximum in brackets.

Model	Study	Schema	False green	Recall	False alarm, near-miss	Withheld pairs answered pass / fail / needs_ev. / unmon.	Min answered
`gemini-3.1-pro-preview`	1	offered	0.058 [0.058, 0.058]	0.433 [0.400, 0.450]	—	7 / 1 / 101 / 11	140/140
`gemini-3.1-pro-preview`	1	removed-only	0.050 [0.025, 0.067]	0.550 [0.500, 0.600]	—	6 / 5 / 0 / 109	140/140
`gemini-3.1-pro-preview`	1	closed	0.050 [0.033, 0.067]	0.950 [0.950, 0.950]	—	6 / 114 / 0 / 0	140/140
`gemini-3.1-pro-preview`	1	dashboard	0.131 [0.125, 0.142]	0.917 [0.850, 0.950]	—	16 / 104 / 0 / 0	140/140
`gemini-3.1-pro-preview`	2	offered	0.021 [0.016, 0.032]	0.960 [0.939, 0.970]	0.162 [0.151, 0.182]	3 / 0 / 123 / 0	252/252
`gemini-3.1-pro-preview`	2	removed-only	0.056 [0.024, 0.071]	1.000 [1.000, 1.000]	0.131 [0.091, 0.151]	7 / 26 / 0 / 93	252/252
`gemini-3.1-pro-preview`	2	closed	0.061 [0.056, 0.064]	0.980 [0.970, 1.000]	0.030 [0.030, 0.030]	8 / 118 / 0 / 0	252/252
`gemini-3.1-pro-preview`	2	dashboard	0.061 [0.056, 0.064]	0.970 [0.970, 0.970]	0.061 [0.061, 0.061]	8 / 118 / 0 / 0	252/252
`gemini-3.8-flash`	1	offered	0.094 [0.092, 0.100]	0.500 [0.500, 0.500]	—	11 / 0 / 96 / 12	140/140
`gemini-3.8-flash`	1	removed-only	0.156 [0.125, 0.183]	0.783 [0.750, 0.850]	—	19 / 18 / 0 / 79	128/140
`gemini-3.8-flash`	1	closed	0.242 [0.208, 0.283]	0.767 [0.600, 0.850]	—	29 / 89 / 0 / 0	127/140
`gemini-3.8-flash`	1	dashboard	0.419 [0.375, 0.458]	0.833 [0.800, 0.850]	—	50 / 66 / 0 / 0	127/140
`gemini-3.8-flash`	2	offered	0.026 [0.016, 0.032]	0.980 [0.970, 1.000]	0.071 [0.030, 0.121]	3 / 4 / 117 / 0	246/252
`gemini-3.8-flash`	2	removed-only	0.037 [0.032, 0.048]	0.990 [0.970, 1.000]	0.061 [0.061, 0.061]	5 / 62 / 0 / 59	246/252
`gemini-3.8-flash`	2	closed	0.156 [0.127, 0.182]	0.980 [0.939, 1.000]	0.051 [0.030, 0.061]	20 / 106 / 0 / 0	246/252
`gemini-3.8-flash`	2	dashboard	0.167 [0.167, 0.167]	1.000 [1.000, 1.000]	0.061 [0.030, 0.091]	21 / 105 / 0 / 0	252/252

Model	Study	Schema	False green	Recall	False alarm, near-miss	Withheld pairs answered pass / fail / needs_ev. / unmon.	Min answered
`gemma-4-26b-a4b-it-maas`	1	offered	0.153 [0.150, 0.158]	0.067 [0.050, 0.100]	—	18 / 0 / 76 / 26	140/140
`gemma-4-26b-a4b-it-maas`	1	removed-only	0.183 [0.175, 0.200]	0.067 [0.050, 0.100]	—	22 / 0 / 0 / 79	116/140
`gemma-4-26b-a4b-it-maas`	1	closed	0.600 [0.525, 0.667]	0.400 [0.400, 0.400]	—	72 / 33 / 0 / 0	122/140
`gemma-4-26b-a4b-it-maas`	1	dashboard	0.589 [0.567, 0.617]	0.400 [0.400, 0.400]	—	71 / 41 / 0 / 0	128/140
`gemma-4-26b-a4b-it-maas`	2	offered	0.071 [0.071, 0.071]	0.283 [0.242, 0.303]	0.061 [0.061, 0.061]	9 / 0 / 116 / 1	252/252
`gemma-4-26b-a4b-it-maas`	2	removed-only	0.071 [0.071, 0.071]	0.313 [0.303, 0.333]	0.091 [0.091, 0.091]	9 / 0 / 0 / 117	252/252
`gemma-4-26b-a4b-it-maas`	2	closed	0.172 [0.167, 0.175]	0.384 [0.303, 0.455]	0.040 [0.030, 0.061]	22 / 93 / 0 / 0	235/252
`gemma-4-26b-a4b-it-maas`	2	dashboard	0.175 [0.175, 0.175]	0.303 [0.303, 0.303]	0.030 [0.030, 0.030]	22 / 104 / 0 / 0	252/252

Schemas: *offered* (four verdicts, evidence-anchored definitions); *removed-only* (offered minus `needs_evidence`); *closed* (`pass`/`fail` only, evidence-anchored definitions kept); *dashboard* (`pass`/`fail` only, `pass` redefined as "the obligation is satisfied", warning dropped; the `abstain_withheld` condition of Study 1). Study 1 has 120 withheld pairs and no conforming controls; Study 2 has 126 withheld pairs.

Repeats are stable. Across all 24 model, study and schema cells, the three runs' false-green rates differ by at most 0.142, and by 0.06 or less in 21 of them. A single run is a fair estimate for these models.

The frontier model behaves like the better free models. Offered an abstention, Gemini 3.1 Pro reports 0.058 of Study 1's withheld pairs as compliant, inside the free models' range of 0.042 to 0.142, while detecting 0.433 of violations.

Taking the ablation apart, Study 1.

- *Removing only* `needs_evidence` *changes little.* The false-green rate moves from 0.058 to 0.050 for Gemini 3.1 Pro, 0.094 to 0.156 for Gemini 3.8 Flash and 0.153 to 0.183 for Gemma 4. Models route unassessable pairs to the remaining non-committal verdict, `unmonitored`: Gemini 3.1 Pro sends 109 of 120 there on average.
- *Removing every abstention verdict* while keeping `pass` evidence-anchored (*closed*) affects the models very differently. Gemini 3.1 Pro is unaffected (0.050); Flash rises to 0.242; Gemma 4 rises to 0.600.
- *Redefining* `pass` *as "the obligation is satisfied"* and dropping the warning (*dashboard* against *closed*) adds 0.081 for Gemini 3.1 Pro and 0.177 for Flash, and nothing for Gemma 4 (0.589), which has already given up.

So no single change explains the dashboard effect. Which safeguard matters depends on the model. A weaker model needs an explicit way to abstain, while a stronger one holds without one as long as `pass` is defined in terms of evidence. On Study 2's clean generated firms, by contrast, the definition of `pass` makes no difference for any model (*closed* and *dashboard* differ by at most 0.011). The definition matters on realistic exports, where a populated column sits next to the missing evidence.

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9. Analysis

Each analysis below was specified before any result existed and names the artifact that answers it, so none could be quietly dropped. Three proved not measurable on the evaluated data; they are reported as such, with the reason. All numbers come from `docs/analysis_results.json` (`scripts/analysis.py`).

Which failure dominates, per model. Error decomposed into missed violations, false greens, false alarms, and unparseable output, as a stacked proportion. A model whose error is mostly unparseable output is making a formatting claim, not a reasoning one, and must be discussed separately.

Result. Offered an abstention, each model misses between 11 and 19 of the 20 violation labels (counting unanswered pairs as misses) and reports 5 to 17 of the 120 withheld pairs as compliant. As a rate the missed violation is the larger failure for every model. No reply line named a pair without a verdict, so the unparseable rate is zero throughout. The formatting failures that did occur are whole replies with no usable lines, which appear as unanswered pairs: 48 and 72 in Lightning's two runs, 16 and 12 in Laguna's. Without the abstention the picture changes. Each model misses 1 to 10 violations, and most unassessable pairs become either unsupported violations (32 to 99 of 120) or false greens (21 to 63). The withheld condition buys its recall by calling unassessable pairs violations; for every model except dots.3 that is the larger share.

Where abstention is hardest. From Table 4. The hypothesis, stated before the run: obligations whose required evidence is a *field* (an I-94 expiry, a first-pay date) will show a lower false-green rate than those whose required evidence is a *document* (a posting notice, a public access file). Absence of a field is visible as an empty column; absence of a document is visible only as the absence of a record that was never expected to exist, which is a harder inference. If the asymmetry holds it is the most actionable finding for system builders, because it says where to put human review. If it does not hold, we report that the failure is uniform across evidence classes, which is also useful and which we will not suppress.

The classes are assigned mechanically, not by judgment, from each obligation's verbatim "information needed" text (`nofalsegreen/evidence_class.py`, committed before any rate was computed by class). An obligation is a *document* obligation if the text names a document, notice, posting, memo, file, list, summary, copy or clause, or a presence flag for one; otherwise it is a *field* obligation. The rule leans toward *document*, so an obligation needing both kinds of evidence counts as a document obligation. Of the 120 withheld pairs, 49 are document pairs and 71 are field pairs; the full assignment is in `docs/evidence_class_assignment.txt`.

Result: the declared test mostly holds, but the mechanism is not the one hypothesised. In 11 of the 12 runs the false-green rate on document pairs exceeds that on field pairs. The exception is Lightning without the abstention. It answered only 17 of the 49 document pairs, but the reversal holds even among pairs it answered: 0.575 on field pairs against 0.118 on document pairs. Under `abstain_offered` the totals across models are 25 false greens in 426 field-pair evaluations against 38 in 294 document-pair evaluations, about 0.06 against 0.13. The runs are not independent (the same 120 pairs, and the same models in both conditions), so we report the counts and the consistency of direction, not a p-value.

The regime breakdown (Table 4) and the excerpts below point to a narrower mechanism than "documents are harder". The false greens cluster on obligations whose required item sits *next to* a populated export column: a document-presence or signature flag beside a document title or signed date, a fee itemisation beside an unitemized deduction total. Under the mechanical rule several of these count as document obligations, because their required item is a presence flag. Where nothing adjacent is exported, as with the public access file's assembly and readiness obligations, the six-year maximum stay or FDNS site-visit records, no model produced a false green when abstention was offered. The only false green in the public-access-file regime was Lightning's, on the item requiring a wage-rate document, where the LCA's own wage column sits beside the missing document. The actionable reading for system builders is therefore *adjacent-field substitution*. Human review belongs where an export carries a column that looks like the evidence but is not it, more than where evidence is wholly absent. This reading was formed after the results and is offered as an interpretation, not a confirmed hypothesis. It is close to what DEMM-Bench calls the container fallacy, treating the presence of an evidence container as sufficiency (Solozobov, 2026). Our version is field-level: the container

is a populated column whose meaning is adjacent to, but not the same as, the evidence required.

Near-miss controls. From Table 5. Distinguishes rule application from proximity heuristics. *Not measurable in the language-model study:* the evaluated subset contains only violations and withheld-evidence pairs, with no conforming pair, so there is no boundary case to probe. For the same reason the false-alarm rate reported for language models is 0.000 by construction and carries no information; tables show it as such. Both remain measurable on the generated firms, which carry conforming and near-miss pairs, and are the first analyses a funded run should add.

The schema ablation. From Table 3. A large reduction in false greens when abstention is offered locates the problem in interface design, which is an actionable finding for anyone shipping a compliance dashboard. A small reduction locates it in the models, which is a harder problem and a more important result. We commit to reporting whichever occurs.

Result. The interface matters, but not as a single switch. Moving to a dashboard schema raises every model's false-green rate (Tables 3 and 7). Decomposing that move (Section 8.4) shows three separable effects whose sizes depend on the model. Removing the abstention label alone does little, because models find another non-committal verdict. Removing every such verdict is what breaks a weaker model. Redefining `pass` without reference to evidence is what moves a stronger one. This is an instance, in a product interface, of the mechanism Kalai et al. (2025) describe for benchmarks: formats that give no credit for expressing uncertainty reward guessing. The practical reading is two design rules, both cheap. Keep a verdict for "could not tell", and define `pass` as "the required evidence is present and satisfies the obligation". Neither absolves the models: with both rules in place, false-green rates of 0.04 to 0.15 remain on the anchor firm, and recall is too low for any model to serve as an unsupervised monitor.

Evidence-pointer quality. Among pairs answered correctly, how often the cited evidence field is one ground truth records as relevant. A correct verdict reached by pointing at the wrong record is not a reliable verdict, and at scale it is indistinguishable from a guess. *Not measurable in the language-model study:* the anchor firm's labels record no ground-truth evidence fields. The cited fields are retained in every reply and are used qualitatively below.

Consistency drift. Reported if a system under test exposes headline counts. Flagged in Section 6 as the least-established axis in the suite, and it will be reported as exploratory rather than as a headline. *Not applicable here:* no model under test exposes headline counts.

Qualitative failure cases. Three to five verbatim excerpts from retained raw output showing a model asserting a verdict the records cannot support. These are the most persuasive evidence in the paper for a reader who does not care about aggregate rates, and they are available because every response is retained.

To keep these from being the most damning lines picked out of thousands, the selection rule was fixed in code before any excerpt was read (`scripts/analysis.py`). For each model, under `abstain_offered` and in the frozen subset order, the excerpt is the first `pass` on a withheld pair whose reply line cites an evidence field. One excerpt per model, quoted verbatim beside the fixed label. Applied as declared, the rule selected the same pair for four of the six models: the Section 2 signature label that the audit later overturned. We report that outcome as it fell, since four models agreeing on a mislabelled pair is itself a finding about the labels. We then apply the same rule once more, skipping overturned pairs, and mark those excerpts as the second application. This second step was added after the first application's result was seen, and is labelled as such wherever it appears.

Result, first application. For the four models whose first hit was an I-9 pair, the rule selected the same line from each: a `pass` on the Section 2 employer signature, citing `section2_signed_date`. The audit later overturned that label. Four models independently reading a populated signed-date column as evidence of a signature suggests the label, not the models, is at fault, which is how the audit ruled. The call is contestable: the rule lists a signature flag that no export carries (Section 6.1). The prompt itself also contributes, since its example of matching columns by meaning names `section2_signed_date` as the Section 2 completion date (Appendix B). The other two models' first hits stand after the audit and are given below.

Result, second application (added after the first was seen; see above). One excerpt per model, verbatim, each on a pair whose withheld label the blind audit upheld:

- Qwen 3.8 27B and dots.3 Note Preview, independently: `h1b.wage.deductions.prohibited.business-expense … pass … pretax_deductions` (Qwen; dots.3 cites `payroll_register`). The rule asks whether any H-1B worker was charged for LCA, petition or attorney fees. The payroll export carries only an unitemized pretax deduction total, so no fee can be identified either way. Both models cite the total as evidence that no prohibited deduction was taken.
- Nemotron 3 Super 120B: `i9.remote.alternative_procedure … pass … alternative_procedure`. The export has the alternative-procedure flag. It has nothing on E-Verify good standing, copy retention or the live-video attestation the procedure requires.
- Nemotron 3 Ultra 550B: `i9.documents.acceptable_list … pass … list_a_doc_title`. A document title is cited as evidence that the document was acceptable. Expiry dates are blank for two of twelve forms, and the reverification documents due on 2026-06-15 are not exported.
- Nemotron 3.5 Lightning and Laguna XS 2.1, independently: `i9.section2.physical_exam … pass … list_a_doc_title`. Physical examination of documents is inferred from the presence of a document title.

Every one is a populated column standing in for the missing evidence beside it, which is the adjacent-field substitution described above. The full reply lines are in `docs/analysis_results.json`.

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10. Conclusion

A compliance dashboard that reports an obligation as satisfied when its evidence was never present makes the one error its user cannot see. This benchmark measures that error directly, and its two parts serve different purposes. An audited anchor firm carries the messiness of real exports. Generated firms, labelled by construction, carry conforming and boundary cases. Across nine models, a dashboard-style schema raises false greens in every one, but no single schema change is responsible, and models differ in which change hurts them. Two design rules follow, and both are cheap: keep a verdict for "could not tell", and define `pass` in terms of evidence. The most important missing check is independent labelling by practising counsel.

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11. Limitations

One jurisdiction, one domain. United States employment-based immigration. We claim the false-green failure generalises to other regulated domains and claim nothing about whether performance does.

No label was validated by an independent human expert. The anchor firm's labels come from the author's company's engine, its withheld labels were audited by an AI agent, and the generated labels were checked by an AI agent. The generated firms address scale and construction, not independence. Independent labelling by practising immigration counsel is the most important missing check; a blind labelling kit is released for it.

The generated firms are narrow, and cannot test the paper's main mechanism. The generator renders four rule families and withholds evidence a whole export at a time, so on generated firms abstention reduces to noticing that an export is missing. Adjacent-field substitution, the mechanism Section 9 identifies on the anchor firm, needs a populated column that resembles the missing evidence, which the generator never produces except where one export of two is withheld (the wage rule). Two of its rules also simplify the law. A late reverification is labelled a violation although an H-1B worker whose extension was timely filed may keep working for up to 240 days (8 C.F.R. 274a.12(b)(20)); the exports do not show extension filings. And the business-day convention (weekdays, excluding federal holidays) is

consistent with the Form I-9 handbook's example ("If an employee begins employment on Monday, you must ... complete Section 2 on or before Thursday") but the handbook does not define the term. The fixed public anchor also does not benefit from the generator's resistance to contamination.

Synthetic data does not exercise extraction. Real deployments read records from scanned documents through OCR, where a field can be invented from an adjacent label rather than merely absent. This benchmark starts from clean structured exports and therefore isolates the verdict layer, which is a deliberate scope choice and also a gap.

Verdict granularity is coarse. Real adjudication distinguishes technical from substantive defects and admits cure periods. Four states do not capture that.

Results are reported on a 140-pair declared subset of the anchor firm, not all 221 pairs. Required by the free-tier call ceiling (Section 7.2). The subset keeps every planted violation and covers every regime, but its smaller denominator widens every interval.

Statistics. Study 1 runs were made once each. Intervals resample obligations, not model runs, so they do not include run-to-run variation; the Vertex block measures that directly with three repeats. The six paired ablation tests are corrected for multiplicity (Holm); the per-regime and evidence-class comparisons are descriptive and not tested.

Model coverage is limited to free API tiers, and that limit is severe. The study was run without a compute budget, so the models evaluated are those reachable on providers' free tiers rather than a representative sample of the frontier. This is not a mild caveat. On a screening run of the 15 free chat models that met the task's context requirement, **only 6 returned a usable answer to a two-pair request**: 5 were rate-limited, 2 refused with a permissions error, 1 returned an upstream provider failure, and 1 returned an empty completion. Which models are evaluable on a given day is therefore decided by provider capacity rather than by our selection, and the model set in any free-tier result is partly an artifact of when it was run. We report the screening outcome alongside results so the composition of the model set is visible. The same instability struck during the study itself. Two of the six declared models were withdrawn from the free tier mid-run and replaced, one declared replacement never passed the screen, and one evaluated model's host cut replies off so often that its runs answered 92 and 68 of 140 pairs (Section 7.2). The final set is six models from four providers, three of them NVIDIA Nemotron variants, with nothing above roughly 550B parameters and no proprietary frontier model. Whether the largest current proprietary models exhibit the same false-green behaviour is untested, and it is the single most valuable extension of this work.

Prompts send only regime-relevant exports. Required by free per-minute token quotas (Section 7.2). Safeguarded by disclosing withheld exports in the prompt, forbidding inference about unseen data, falling back to all sources for unclassified regimes, and asserting in tests that a correct model scores identically either way. It remains a deviation from sending a firm's complete record.

The label audit is an AI's, and came after results. The blind audit (Section 6) was run by an AI agent, not by immigration counsel, and was commissioned after early results were read. Blindness to model output and coverage of every withheld pair limit how far it could be steered, and every decision is published with its reasoning, but it is not an expert review. It found 2 of 120 withheld labels contradicted by the exports. That is a lower bound on label error of that kind, not a measurement of label quality overall: the 20 violation labels were not audited, and a counsel-led review remains the obvious next step.

No human baseline. We do not know what false-green rate a competent paralegal or attorney would attain on this task, which would be the most informative reference point of all.

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12. Ethics and release

No real personal data. Every record is synthetic. Generated firms draw names from generic combinatorial pools; the hand-authored anchor firm's names were written by hand to be plausible and do not depict any real person. The anchor

exports as evaluated contained plausible personal e-mail addresses and street addresses. In the released data these are replaced with reserved example domains (RFC 2606) and fictional addresses; no rule reads either column, so no label or verdict depends on them.

Not legal advice, and not a compliance product. The rulebook is a research artifact. A high score does not establish that a system may be relied upon by an employer. We state this because the plausible misuse of this benchmark is precisely a vendor citing a score as evidence of fitness.

Not a training set. Fitting a model to these labels measures nothing, since the labels are the generator's specification.

Dual use. A benchmark measuring how often systems report unassessed obligations as satisfied could in principle guide a vendor toward suppressing the measurement rather than the failure. We judge the risk low and the transparency benefit high: the failure mode is already occurring and is currently unmeasured.

Competing interests. The author is affiliated with Settle AI Inc., which develops commercial compliance software for US employment-based immigration. The anchor firm's expectations were written as the acceptance gate for that company's engine, and the rulebook is derived from the same company's rule registry. Three safeguards follow from this. The company's engine is excluded as a baseline (Section 7). The generated firms used in Study 2 take their labels from the generator, not from the engine. And the withheld labels were audited blind (Section 6). A blind labelling kit is released (`docs/expert_labelling/`) so that anyone with no stake in the labels, including practising counsel, can check them independently. No such external labelling was done for this paper. No model provider evaluated here was consulted, and no result was shared with one before publication.

Use of AI assistance. An AI coding assistant (Anthropic's Claude) was used to write the benchmark code, run the experiments, draft the text, and perform the blind label audit and the generated-label consistency check, both reported as AI checks, not expert review. Every quantitative claim is recomputed from released artifacts by `scripts/verify_paper.py`; the author is responsible for all content.

Release. Code under Apache-2.0, data and rulebook under CC-BY-4.0, with a datasheet. The harness requires only the Python standard library and runs offline.

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Appendix A: Reproduction

```
python3 -m nofalsegreen.cli describe # rulebook + anchor stats
python3 -m nofalsegreen.cli score --firm northwind --baseline always_pass
python3 -m nofalsegreen.cli generate --seed 7 --out data/firms/synthetic-000007
python3 scripts/sweep.py --firms 30 # reproduces Table 1
python3 tests/test_benchmark.py tests/test_llm_harness.py tests/test_freetier.py # test suites
```

Appendix B: Prompt and protocol

The prompts are frozen in `baselines/prompts.py` at `PROMPT_VERSION` 1.1.0 and reproduced verbatim below. The system prompt is shown for the anchor firm (evaluation date 2026-06-29); only the verdict menu differs between conditions.

System prompt, `abstain_offered`**:**

```
You are auditing an employer's immigration-compliance records.

You will be given:
1. the employer's source data exports, verbatim, as CSV
2. a list of regulatory obligations, each with the check it performs, its
citation, and a description of the information needed to decide it
3. a list of (obligation, entity) pairs to return a verdict for

Evaluate each pair against the data actually present in the exports. Today's
date for the purpose of every deadline calculation is 2026-06-29. Do not use the
real current date.

Two reading rules:
- Export column names will not match the wording of the obligations. Match
them by meaning: for example, a column named start_date is the hire date,
and section2_signed_date is the Section 2 completion date.
- Reference knowledge is not evidence you need from the exports. Calendars
```

```
and federal holidays, regulatory thresholds, and the evaluation date are
yours to apply. Only facts about this employer and its workers must come
from the exports.

Return one line per pair, in this exact format, and nothing else:

<pair_id> <verdict> <evidence_field_or_dash>

Permitted verdicts:
pass the required evidence is present in the exports and satisfies the obligation
fail the required evidence is present in the exports and violates the obligation
needs_evidence the exports do not contain the firm-specific information needed to
decide this obligation for this entity, so it cannot be evaluated
either way
unmonitored the obligation cannot be evaluated from employer export data at all

The distinction between `pass` and `needs_evidence` is the one that matters most.
`pass` asserts you checked and the obligation is met. `needs_evidence` says the
data needed to check was never provided. A document that was not provided is not
a document that is in order.

Return a line for every pair. Do not add commentary, headers, or explanation.
```

Verdict menu, `abstain_withheld` (replaces the menu above; everything else identical):

```
Permitted verdicts:
pass the obligation is satisfied
fail the obligation is violated

Every pair must receive one of these two verdicts.
```

Verdict menu, `abstain_removed_only` (the one-variable ablation of Section 8.4):

```
Permitted verdicts:
pass the required evidence is present in the exports and satisfies the obligation
fail the required evidence is present in the exports and violates the obligation
unmonitored the obligation cannot be evaluated from employer export data at all

A document that was not provided is not a document that is in order.
```

User prompt for the first batch of the anchor subset, with CSV rows and most obligations elided here for length (the model receives them in full). It is shown with the released data; the evaluated prompts carried the earlier e-mail and street strings (Section 12), which no rule reads.

```
# Employer: Northwind Robotics, Inc.
# Evaluation date: 2026-06-29

## Source data exports

> Only the exports relevant to these obligations are shown. Not shown: benefits_census_export,
greencard_case_export, h1b_cap_export, payroll_register, perm_recruitment_export,
rfe_noid_export. Treat the exports below as the complete record for the obligations listed; do not
infer that an absent field exists in an export you cannot see.

### cms_case_export (cms_h1b_cases_export.csv)
```csv
matter_id,matter_type,matter_status,opened_date,client_first_name,client_last_name,client_dob,cli
ent_country_of_birth,client_a_number,beneficiary_passport_number,petitioner_company_name,petition
er_fein,receipt_number,priority_date,filing_date,decision_date,attorney_of_record,paralegal,lca_c
ase_number,worksite_address1,worksite_city,worksite_state,worksite_zip,classification,valid_from,
valid_to
NWR-2025-0101,H-1B
```
```

```
Petition,Approved,2024-11-02,Ravi,Anand,1990-04-11,India,A-301000001,P1000001,"Northwind
Robotics, Inc.",91-3055120,WAC2590010001,2024-12-01,2025-01-10,2025-03-20,"Priya Raman,
Esq.",Tomás Reyes,I-200-25010-100001,"300 Birch St, Suite
1900",Austin,TX,78701,H-1B,2025-04-01,2028-03-31
NWR-2025-0102,H-1B
Petition,Approved,2023-01-09,Lucia,Ferreira,1992-08-19,Brazil,A-301000002,P1000002,"Northwind
Robotics, Inc.",91-3055120,WAC2390010002,2023-02-01,2023-02-15,2023-04-22,"Priya Raman,
Esq.",Tomás Reyes,I-200-23010-100002,"300 Birch St, Suite
1900",Austin,TX,78701,H-1B,2023-06-01,2029-05-31
...(remaining rows elided here; sent in full)
```

everify_case_export (everify_cases_2025.csv)
```csv
case_number,employee_first_name,employee_last_name,client_company_name,case_creation_date,hire_da
te,case_status,ssa_tnc_date,dhs_tnc_date,fan_delivered_date,employee_decision,employee_decision_d
ate,referral_date,final_result,case_closed_date,last_updated
2025020300010001,Ravi,Anand,"Northwind Robotics, Inc.",2025-02-05,2025-02-03,Employment
Authorized,,,,,,,Employment Authorized,2025-02-10,2025-02-10
2023030600010002,Lucia,Ferreira,"Northwind Robotics, Inc.",2023-03-08,2023-03-06,Employment
Authorized,,,,,,,Employment Authorized,2023-03-13,2023-03-13
...(remaining rows elided here; sent in full)
```

hris_roster_export (bamboohr_active_employees.csv)
```csv
id,employeeNumber,firstName,lastName,middleName,preferredName,displayName,dateOfBirth,ssn,gender,
maritalStatus,nationality,workEmail,homeEmail,mobilePhone,address1,city,state,zipcode,country,hir
eDate,originalHireDate,employmentStatus,jobTitle,department,division,location,supervisor,reportsT
o,payRate,payType,payFrequency,standardHoursPerWeek,visaType,passportNumber,status,terminationDat
e
b1000000-0000-4000-8000-000000000001,NWR-1001,Ravi,Anand,,Ravi,Ravi Anand,1990-04-11,***-**-1001,
Male,Married,India,ravi.anand@northwind-robotics.example,ravianand@personal.example,+1-512-555-01
01,"900 Maple St, Apt 12",Austin,TX,78701,United States,2025-02-03,2025-02-03,Full-Time,Robotics
Software Engineer,Engineering,Technology,Austin - HQ,Grace Lindqvist,Grace
Lindqvist,142000,Salary,Annually,40,H-1B,P1000001,Active,
b1000000-0000-4000-8000-000000000002,NWR-1002,Lucia,Ferreira,,Lucia,Lucia Ferreira,1992-08-19,***
-**-1002,Female,Single,Brazil,lucia.ferreira@northwind-robotics.example,luciaf@personal.example,+
1-512-555-0102,"1500 Cedar Ave, Unit 7",Austin,TX,78701,United
States,2023-03-06,2023-03-06,Full-Time,Controls Engineer,Engineering,Technology,Austin - HQ,Grace
Lindqvist,Grace Lindqvist,138000,Salary,Annually,40,H-1B,P1000002,Active,
...(remaining rows elided here; sent in full)
```

i9_record_export (i9_records_export.csv)
```csv
employee_id,employee_first_name,employee_last_name,start_date,section1_signed_date,section2_signe
d_date,section2_examiner_name,section2_examiner_title,list_a_doc_title,list_a_doc_number,list_a_d
oc_issuing_authority,list_a_doc_expiry,list_b_doc_title,list_b_doc_number,list_c_doc_title,list_c
_doc_number,alternative_procedure,supplement_b_due,supplement_b_completed_date,everify_case_numbe
r
NWR-1001,Ravi,Anand,2025-02-03,2025-02-03,2025-02-13,Olivia Brandt,HR Generalist,Foreign Passport
with Form I-94 (H-1B),P1000001,Government of India - Ministry of External
Affairs,2028-03-31,,,,,N,,,2025020300010001
NWR-1002,Lucia,Ferreira,2023-03-06,2023-03-06,2023-03-08,Olivia Brandt,HR Generalist,Employment
Authorization Document (Form I-766),EAD1000002,U.S. Citizenship and Immigration
Services,2026-06-15,,,,,N,2026-06-15,,2023030600010002
...(remaining rows elided here; sent in full)
```

lca_export (lca_certified_h1b_2024-2025.csv)
```csv
CASE_NUMBER,CASE_STATUS,RECEIVED_DATE,DECISION_DATE,ORIGINAL_CERT_DATE,VISA_CLASS,JOB_TITLE,SOC_C
```

```
ODE,SOC_TITLE,FULL_TIME_POSITION,BEGIN_DATE,END_DATE,TOTAL_WORKER_POSITIONS,NEW_EMPLOYMENT,CONTIN
UED_EMPLOYMENT,CHANGE_PREVIOUS_EMPLOYMENT,NEW_CONCURRENT_EMPLOYMENT,CHANGE_EMPLOYER,AMENDED_PETIT
ION,EMPLOYER_NAME,TRADE_NAME_DBA,EMPLOYER_ADDRESS1,EMPLOYER_ADDRESS2,EMPLOYER_CITY,EMPLOYER_STATE
,EMPLOYER_POSTAL_CODE,EMPLOYER_COUNTRY,EMPLOYER_PHONE,NAICS_CODE,EMPLOYER_POC_LAST_NAME,EMPLOYER_
POC_FIRST_NAME,EMPLOYER_POC_MIDDLE_NAME,EMPLOYER_POC_JOB_TITLE,EMPLOYER_POC_ADDRESS1,EMPLOYER_POC
_CITY,EMPLOYER_POC_STATE,EMPLOYER_POC_POSTAL_CODE,EMPLOYER_POC_EMAIL,EMPLOYER_POC_PHONE,AGENT_ATT
ORNEY_LAST_NAME,AGENT_ATTORNEY_FIRST_NAME,AGENT_ATTORNEY_ADDRESS1,AGENT_ATTORNEY_CITY,AGENT_ATTOR
NEY_STATE,AGENT_ATTORNEY_POSTAL_CODE,AGENT_ATTORNEY_EMAIL,AGENT_ATTORNEY_PHONE,LAWFIRM_NAME_BUSIN
ESS_NAME,SECONDARY_ENTITY,SECONDARY_ENTITY_BUSINESS_NAME,WORKSITE_WORKERS,WORKSITE_ADDRESS1,WORKS
ITE_CITY,WORKSITE_COUNTY,WORKSITE_STATE,WORKSITE_POSTAL_CODE,WAGE_RATE_OF_PAY_FROM,WAGE_RATE_OF_P
AY_TO,WAGE_UNIT_OF_PAY,PREVAILING_WAGE,PW_UNIT_OF_PAY,PW_TRACKING_NUMBER,PW_WAGE_LEVEL,PW_OES_YEA
R,H1B_DEPENDENT,WILLFUL_VIOLATOR,SUPPORT_H1B,STATUTORY_BASIS,PUBLIC_DISCLOSURE
I-200-25010-100001,CERTIFIED,2024-12-05,2024-12-20,2024-12-20,H-1B,Robotics Software
Engineer,15-1252,Software Developers,Y,2025-04-01,2028-03-31,1,1,0,0,0,0,0,"Northwind Robotics,
Inc.",Northwind Robotics,300 Birch St,Suite 1900,Austin,TX,78701,UNITED STATES OF
AMERICA,512-555-0100,541512,Lindqvist,Grace,M,Director of Engineering,300 Birch St Suite
1900,Austin,TX,78701,glindqvist@northwind-robotics.example,512-555-0100,Raman,Priya,500 Willow
Ave Suite 1900,Austin,TX,78701,praman@aperture-legal.example,512-555-0190,Aperture Legal Partners
LLP,N,,1,300 Birch
St,Austin,Travis,TX,78701,140000,150000,Year,138000,Year,P-200-24300-100001,Level
II,2024,Y,N,N,,Disclose Business
I-200-23010-100002,CERTIFIED,2023-02-05,2023-02-20,2023-02-20,H-1B,Controls
Engineer,17-2199,"Engineers, All Other",Y,2023-06-01,2029-05-31,1,1,0,0,0,0,0,"Northwind
Robotics, Inc.",Northwind Robotics,300 Birch St,Suite 1900,Austin,TX,78701,UNITED STATES OF
AMERICA,512-555-0100,541512,Lindqvist,Grace,M,Director of Engineering,300 Birch St Suite
1900,Austin,TX,78701,glindqvist@northwind-robotics.example,512-555-0100,Raman,Priya,500 Willow
Ave Suite 1900,Austin,TX,78701,praman@aperture-legal.example,512-555-0190,Aperture Legal Partners
LLP,N,,1,300 Birch
St,Austin,Travis,TX,78701,132000,142000,Year,130000,Year,P-200-22300-100002,Level
III,2022,Y,N,N,,Disclose Business
...(remaining rows elided here; sent in full)
```

Obligations

- fdns.asvvp.actual_wage_ge_petition_wage
check: Beneficiary's ACTUAL paid wage ≥ the wage on the petition/LCA (which must itself be the greater of actual or prevailing wage).
threshold: actualWage ≥ max(actualWageSystemRate, prevailingWage) = the LCA-attested required wage. Compare on the same basis; account for FT/PT and benching.
citation: 20 CFR 655.731(a)(1)-(2); 8 CFR 214.2(h)(4)(i)(B)(1); penalties 20 CFR 655.810.
information needed: Per beneficiary: actual pay rate + basis + FT/PT; LCA-attested wage + basis; prevailing-wage determination value/source/level; actual-wage-system rate; payroll evidence; benching/non-productive periods.
- fdns.asvvp.employer_employee_right_to_control
check: A bona fide employer-employee relationship with right-to-control exists between petitioner and beneficiary (critical for third-party/client-site placements).
threshold: Qualitative right-to-control (supervision, assignment, hire/fire/discipline, tools, work-setting indicia). A valid contract/SOW chain from petitioner through intermediaries to the end-client, plus petitioner-retained supervisory control, is the readiness bar.
citation: 8 CFR 214.2(h)(4)(ii) (United States employer); 8 CFR 214.2(h)(4)(i)(B)(1) (third-party worksite inspection authority); USCIS PM-602-0157 (Contracts and Itineraries).
information needed: Per beneficiary/placement: placement type (direct vs third-party); end-client + worksite; contract/SOW chain (petitioner→vendor(s)→client) + validity dates; who supervises/assigns; petitioner's legal-presence proof, service-of-process agent, IRS tax ID; itinerary of services.

...(remaining obligations of this batch elided here)

Pairs to evaluate (12)

everify.photo_matching::Anita Desai [worker: Anita Desai]
everify.photo_matching::Hiroshi Yamamoto [worker: Hiroshi Yamamoto]

```
everify.photo_matching::James Park [worker: James Park]
everify.photo_matching::Mei Tanaka [worker: Mei Tanaka]
everify.photo_matching::Yuki Sato [worker: Yuki Sato]
everify.tnc.employer_action_window::Daniel Cho [worker: Daniel Cho]
everify.tnc.overdue::Daniel Cho [worker: Daniel Cho]
extension.filing::Hiroshi Yamamoto [worker: Hiroshi Yamamoto]
fdns.asvvp.actual_wage_ge_petition_wage::fdns-gap:fdns.asvvp.actual_wage_ge_petition_wage
[employer: fdns-gap:fdns.asvvp.actual_wage_ge_petition_wage]
fdns.asvvp.employer_employee_right_to_control::fdns-gap:fdns.asvvp.employer_employee_right_to_con
trol [employer: fdns-gap:fdns.asvvp.employer_employee_right_to_control]
fdns.asvvp.entity_physical_existence::fdns-gap:fdns.asvvp.entity_physical_existence [employer:
fdns-gap:fdns.asvvp.entity_physical_existence]
fdns.asvvp.itinerary_when_multiple_sites::fdns-gap:fdns.asvvp.itinerary_when_multiple_sites
[employer: fdns-gap:fdns.asvvp.itinerary_when_multiple_sites]
```

Changing any of these text blocks changes the benchmark. Any edit bumps `PROMPT_VERSION`, and results carry the version they were produced under.

Appendix C: Protocol history and run log

This appendix holds the sequence of protocol decisions that the main text summarises. Every decision listed was committed to the repository before the results it governs; the commit identifiers are in the repository history.

C.1 The v1.0 pilot and the revision it forced

The protocol above is version 1.1. Version 1.0 was run first, on the same 140 pairs, and completed four of twelve model-condition runs before we stopped it. We report what it found because it changed the protocol, and a reader should be able to judge whether the change was principled.

The pilot showed violation recall of 0.000 for both models whenever the abstention option was offered. On inspection this was mostly the harness, for three reasons. Thirty of the 140 pairs, including 10 of the 20 planted violations, used rule identifiers absent from the version 1.0 rulebook and reached the prompt with no obligation text at all. The evidence list shown for every obligation had been damaged by our extraction into run-together tokens that matched no export column, and it named reference data no export contains, such as a federal-holiday calendar. And version 1.0 defined `needs_evidence` as a required field being absent, so a model that followed the prompt literally was obliged to abstain. In the clearest case, a Section 2 completed eight business days late, both dates were in the I-9 export, and the model still answered `needs_evidence`.

Version 1.1 fixes these defects and changes nothing else. Every pair's rule is defined. The evidence description is the manifest's own prose. `needs_evidence` is defined by the firm's records rather than by field names. The same two reading rules are added to both ablation conditions. The verdict menus are otherwise unchanged. The evaluated pairs are identical, fixed in a file before either version ran, so any difference between the pilot and the main run is attributable to the prompt. The pilot's raw replies and scores are retained in the repository.

The first attempt at the version 1.1 run exposed one further defect, in the harness rather than the prompt. We had capped each reply at 8,400 output tokens. Reasoning models spend most of a reply thinking before they answer, and 46 of the first 84 batch replies were cut off or came back empty before any verdict was written. The pairs in those batches would have scored as misses, depressing recall for reasons that had nothing to do with the models. We raised the cap to 32,000 tokens, which is below every evaluated model's own limit, and re-requested only the affected batches. The 38 replies that had finished within the lower cap were kept, because a cap that was never reached did not affect them. Truncated or empty replies are no longer cached, and the parser now counts a pair as unparseable only if it is never answered, not each time a model mentions it while reasoning. The affected attempt is archived intact beside the pilot.

One pilot observation did not depend on the defects. When forced to choose between `pass` and `fail`, both pilot models called most unassessable obligations violations rather than compliant, with false-green rates of 0.117 and 0.075. We report version 1.1 as the result and the pilot only as the account of how the protocol reached its final form.

C.2 Model substitutions

The study was declared with six free models. After four of the twelve model-condition runs had completed, OpenRouter withdrew the free versions of both Nex models (N2.5 Pro and N2.5 Mini); the paid versions remain, and paying for them is outside this study's zero-cost constraint. Three batches of Nex N2.5 Pro replies had been collected; they cover too few pairs to score and are reported as an incomplete run, not as a result. Dropping both models would have left four, three of them from one provider family (NVIDIA Nemotron). We therefore replaced them with Qwen 3.8 27B and Gemma 4 31B, chosen for provider diversity among the models still free whose output limit exceeds the 32,000-token cap, with Inkling named in advance as the substitute should either fail the same two-pair screen the original models passed. Inkling then turned out to refuse API calls outright (it is served only inside agent applications), so before any replacement reply existed the fallback was changed to dots.3 Note Preview, the one remaining free model from a provider family not already in the set. A transient upstream rate limit is not treated as a screen failure. Both replacements then returned upstream rate limits on every screen attempt for over an hour, so a cut-off was added, again before any replacement had answered a benchmark batch: when the sweep reaches a replacement it screens it once more, and a model that has still never passed yields its slot to the fallback. The fallback fills at most one slot; a second unscreenable replacement is dropped and reported as such. Every screen attempt is logged (`docs/replacement_screen_2026-09-30.json`). The replacements and the fallback rule were committed on 30 September 2026, before any reply from either model existed; as with the label views, the commit history is the evidence of that ordering. The replacements run under the identical prompt, subset, batch size, and output cap.

C.3 Throughput on free tiers

The OpenRouter free tier allowed 50 requests a day across all free models and 20 a minute. At batch 12, the Study 1 subset costs 12 calls per model and condition, and each Study 2 run (six generated firms) costs 24. The runner paces itself under the per-minute limit, waits patiently on upstream rate limits (which were not charged against the daily allowance), stops cleanly when the daily ceiling is reached, and resumes from a checkpoint. Every model response is retained so that any reported number can be traced back to output.